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Deciphering the U.S. metropolitan house price dynamics

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Abstract

In this article, we propose a novel estimator that builds on recent advances in heterogenous estimators to introduce the concepts of cross-sectional heterogeneity and cross-sectional dependency in the machine learning (ML) literature. The performance of the proposed method is evaluated in forecasting house prices at the county level for the 56 most populated Metropolitan Statistical Areas in the U.S., identifying bubbles in local house markets as they form and measuring the returns on a trading strategy based on model's forecasts. In doing so, we find that the proposed method achieves an out-of-sample error of 0.252 in house prices forecasting, while the most accurate econometric estimator has a forecasting error of 0.678 and the most accurate ML 0.763. In terms of bubble identification, the proposed model achieves a 0.470 recall against 0.390 and 0.380 of the most accurate econometric and ML, respectively. Finally, in terms of economic significance, a diversified portfolio of Real Estate Investment Trust stocks achieves an averaged return of 13.1%, which is twice as large as the second most profitable trading strategy. Our work has direct policy implications to market participants and monetary policy authorities as it shapes a new local approach to monitoring the real estate market.

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KEYWORDS
machine learning, house prices, U.S. real estate market

JEL CLASSIFICATION
C23, C45, R32

1 | INTRODUCTION

It is widely recognized that one of the key findings of the Great Recession of 2008 is the need to monitor the housing market closely, as it is a potential determinant of macroeconomic and microeconomic activity (Campbell & Cocco, 2007; Cesa-Bianchi, 2013). House ownership accounts for a large share of households' aggregate wealth (Chetty et al., 2017) and large deviations from "fundamental" values (Stiglitz, 1990), informally called "bubbles," can potentially contribute to financial and economic crises. As Leamer (2007) puts it, "Housing is the business cycle," given that residential investment typically acts as an early-warning indicator on the eve of a recession; is a canary in the coal mine. In addition to contributing to a general decline in the economy, the abrupt burst of a bubble has substantial heterogenous negative economic effects across households and neighborhoods. In hindsight, efforts to avoid the formation of bubbles would be sound public policy (Crowe et al., 2013). There is, however, a lack of consensus in the literature regarding the methodology to measure the fundamental house prices and any potential discrepancies with actual house prices. Methodology choice and analysis get even more complex when it comes to house bubble forecasting, specifically at the local level before the formation spillovers nation-wide.

The benefit from accurate forecasting of house prices at a subnational level is twofold. First, as urban economic theory and empirical evidence suggest, the housing markets are local in nature and exhibit a high level of spatial heterogeneity. Thus, policy suggestions on averting bubble formations should be tailor-made to local housing markets that spur regional, although highly contagious, bubbles. Moreover, the direct and indirect effects of real estate investments and related spillovers are of key interest for the development in stagnating or declining city-regions (Buchholz & Bathelt, 2021). To this end, recent applications of panel data estimators illustrate heterogenous housing wealth effects, housing demand and supply elasticities, and consumption dynamics at both the regional and the inter-regional level (Holly et al., 2010; Lai & Van Order, 2017; Oikarinen & Engblom, 2016; Oikarinen et al., 2018; Paciorek, 2013).

The heterogeneity of house price dynamics and spatial heterogeneity at the regional level is a well-established notion in the econometric literature, since the pioneering work of Case and Shiller (1990) and Lai and Van Order (2017). The latter develop a price model based on the Gordon dividend discount approach and use the mean group (MG) (Pesaran & Smith, 1995) and the pooled MG estimator (Pesaran et al., 1999) to study price momentum and reversals across 45 U.S. Metropolitan Statistical Areas (MSA), finding little heterogeneity in the long-run, but notable short-term deviations of price dynamics across regions. Oikarinen et al. (2018) expand the previous study, testing for spatial differences across 70 U.S. MSAs using additional heterogenous panel estimators (Chudik & Pesaran, 2015; Pesaran, 2006). The authors argue that short-term momentum and reversion dynamics show substantial spatial heterogeneity, while the long-term income elasticity is higher in areas of low (inelastic) supply.

Aquaro et al. (2021) introduce heterogeneous spatial lag coefficients on house prices, using a novel quasi maximum likelihood estimation method on a sample of 338 U.S. MSAs that supports

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a significant degree of spatio-temporal heterogeneity, reaching to similar conclusions regarding the locality of housing markets. In a recent study, Koeniger et al. (2022) extend the analysis to Germany, Italy, and Switzerland, reaching to comparable results to the U.S. market regarding heterogeneity. The detected differences in the pass-through of monetary policy shocks to mortgage rates, housing tenure transitions, rents, and house prices indicate that differences in institutional features such as indexing rents to mortgage rates, characteristics of the mortgage market and the extent of public housing shape the transmission channel of monetary policy to the housing market.

On a different strand, the relevant literature explores the ability to detect bubble formation. Since the seminal article of Blanchard and Watson (1982), the notion of bubbles has been studied exhaustively, while all central monetary authorities monitor closely bubble formations. As suggested by Phillips and Yu (2011), bubbles can be detected in the U.S. home market for the period 2002 to 2007 by simply using forward—recursive Dickey–Fuller tests. On a similar framework, Fabozzi and Xiao (2019) provide empirical evidence on home price bubble formations in Miami, Chicago and New York for the period January, 1999 – October, 2006. Bourassa et al. (2019) investigate several methods for detecting bubbles in house prices, using data for six major cities (Helsinki—Finland, Geneva and Zurich—Switzerland, and Chicago, Miami, and San Francisco—U.S.) and find that the rental-adjusted price (the simple price-to-rent ratio) serves as the best indicator to identify and measure a real estate bubble.

Among the few studies that depart from highly aggregated data, Begley et al. (2019) use micro data to the individual house price level and find that the price–rent ratios evolve similarly across equivalent property types, but at substantially different growth rates across regions (U.S. state), with the main driver of heterogeneity being market agents' expectations. Nevertheless, none of the aforementioned studies includes the notion of cross-correlation in their examination framework, since they examine each region independently (univariate modeling).

Along with the advances in econometrics, the last two decades have encouraged the use of machine learning (ML) methods in economics.[1] The motivation in the use of ML stems from the fact that typical econometric approaches aim at estimating generalized population average and subject-specific effects, making assumptions on the nature of these effects (i.e., fixed effects, random effects, heterogenous panel models, etc.) and testing them statistically (endogenous validity). In contrast, ML methods focus on methods (algorithms) for conditional predictions, with the emphasis shifted to the generalization ability of the model to new populations (exogenous validity). Overall, econometric models are better suited to problems when the statistical properties of the estimators are of interest, while the ML family of estimators focuses on the generalization ability and is thus better suited to prediction tasks (Mullainathan & Spiess, 2017).

The applications of ML can be divided into two main approaches; one that focuses on model selection to address under-identification issues when the number of covariates relative to the available observations can be large and another that comprises to cross-validation methods using iterative training schemes, treating problems of overfitting. Representative of the first strand of literature is the seminal article of Mullainathan and Spiess (2017), where the authors forecast 41,808 house values of owner-occupied units from the 2011 American housing survey, considering 150 covariates including unit characteristics and quality measures in a "horse-race" between different methods. The forecasting task evaluates ordinary least squares regression against a variety of ML techniques, concluding that the random forests (RF) method (a bootstrapped version of a decision trees estimator) outperforms only marginally the least squares regression.

[1] For a more detailed exposition, see among others Athey and Imbens (2019) and Gu et al. (2020).

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A representative example of the second approach is Plakandaras et al. (2015), in which the authors forecast the S&P/CS U.S. house prices index at the national level over the period 1890–2012, proposing a novel ML method that combines signal processing to the ML Support Vector Regression (SVR) method. The proposed approach clearly outperforms the most accurate in forecasting Bayesian vector autoregressive and Bayesian autoregressive methods. Milunovich (2020) considers 47 different ML algorithms to evaluate Australian real house prices and compares their out-of-sample predictions accuracy over alternative horizons and estimators. He finds that the linear SVR estimator outperforms all other ML algorithms.

To this end, the novelty of this study is to introduce econometric concepts to the ML approach, by evolving the functional form of the popular SVR model from a univariate to a panel estimator, while taking into account heterogeneity among cross-sections. Our work departs substantially from the typical SVR model, exploiting the heterogenous structure of an econometric estimator. In a typical heterogenous econometric estimator, one would allow for individual slope coefficients per cross-section and cluster the variance-covariance matrix to obtain robust standard errors. However, by introducing, as we do in this article, a heterogenous trend parameter (individual per cross-section) both to the functional form of the SVR model as well as the kernel function, we cater for both heterogeneity and cross-correlation in the data. The approach follows closely the econometric literature in testing for heterogeneity effects that vary with time as in Cai (2007) and Chen (2019). The main difference is that the proposed method coded "panel SVR" determines endogenously potential heterogeneity through the kernel matrix.

The proposed panel SVR model is evaluated in forecasting monthly house prices at the county level for the 56 most populated U.S. MSAs (431 counties) for the period January 2000 to May 2022. The sample includes more than half (55%) of the entire U.S. population and more than 66% of the population residing in urban areas,[2] suggesting that our sample is directly representative of the entire U.S. population. Its time span includes the subprime mortgage crisis period, the sluggish recovery of the 2010s and the COVID-19 pandemic that all exhibit significant fluctuations in house price dynamics. The dissimilarities between MSAs are observed in terms of the number of counties included, income, number of housing units, construction costs, and house prices.

The forecasting performance of the panel SVR model is evaluated in a rolling window training scheme of 60 observations (5 years) to account for structural breaks in the dataset. The empirical findings suggest that the panel SVR model exhibits half or less (Root Mean Square Error [RMSE]: 0.296) of the forecasting error of the second most accurate RF model (RMSE: 0.595) in out-of-sample forecasting at the 1-month forecasting horizon, while it maintains its superiority up to the 24-month ahead forecasting horizon (0.240 vs. 0.687). For comparison reasons, we also forecast house prices at the county level treating each MSA as an individual panel and average the error per forecasting horizon. Again, the superiority of the proposed method is highlighted with a RMSE at the short monthly horizon of 1.103 versus a RMSE of 3.257 of the second most accurate model and a RMSE of 1.139 versus 3.411 at the longer 2-year horizon. Thus, accounting for spatial heterogeneity and cross-dependence greatly improves the forecasting accuracy of the selected ML approach.

As a second exercise, we evaluate the ability of the panel SVR model to foresee bubble formations in the same sample of MSAs, denoted as deviations of the forecasted values from their long-run fundamental prices. Compared with the rent–price ratio, the panel SVR model achieves a recall rate of 0.512 on the short monthly horizon against a recall rate of 0.373 for the random walk (RW) model, and a recall rate of 0.455 at the longer 24-month horizon against a recall rate of 0.396 for the second-best MG estimator. Timely identification of a house price bubble has direct policy

[2] See the Supplementary File for more information on the population per MSA.

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implications, as it can be an early indication to avoid further house price inflationary pressure that could eventually contribute to an economic crisis (Martin, 2010, Shiller, 2008).

Nevertheless, while forecasting superiority based on statistical metrics could provide a strong indication that the proposed model can be a useful instrument for market professionals and policy authorities, the true interest lies with the ability to translate forecasts into profitable trading strategies. In doing so, we evaluate the economic performance of forecasts in Real Estate Investment Trusts (REIT) management; trusts that own, operate, or finance income-producing real estate units. Gathering monthly returns on 184 residential REITs active in 40 states that are publicly traded on the stock market for the period January 2000 to May 2022, we access the ability to shape profitable REIT stock portfolios. The empirical findings suggest that the panel SVR model achieves an annualized return of 13.7% at the monthly forecasting horizon, over a return of 7.7% of the artificial neural network (ANN), the model shaping the second most profitable trading strategy at the same horizon. Extending the forecasting horizon to the 24-month period, the panel SVR model achieves an annualized return of 13.1% over a return of 7.1% for the second most profitable SVR model.

Overall, the main motivation of this study is to examine the potential gain in introducing a functional form that accounts for the cross-dependency to the popular SVR model. In an effort to highlight the merits of a panel examination, we select several datasets from the real estate market, as housing prices are local in nature, highly heterogenous and also highly interdependent. To the best of our knowledge, this study is the first application of a panel ML method to economics and one of the very few that introduces modern econometric methods that cater for spatial heterogeneity to house prices bubble identification. In the Supplementary File to this article, the interested reader can find a forecasting exercise of house prices down to the ZIP level, with comparable quantitative results. Our examination does not reveal the large discrepancies in the forecasting performance between econometric and ML methods reported in the literature. Presumably, this difference stems from the use of heterogenous econometric estimators that correct for spatial heterogeneity, largely ignored in previous studies. Finally, the proposed methodology has direct policy implications for regulating authorities that monitor price overvaluations, professionals of the real estate market that seek future trends of the market and REIT managers that trade on stock markets. Harnessing the forecasting ability of panel ML models could reveal future economic disturbances originating from the housing market, while being of real value to market participants.

The remainder of the article is organized as follows: Section 2 presents the data and the methodology used, Section 3 refers to the empirical findings of the article, and Section 4 concludes the study.

2 | DATA AND METHODOLOGY

2.1 | Model structure

The structure of the model in our forecasting exercise is based on a typical stock-flow model of the housing market. According to economic theory, the observed house prices are determined by demand and supply in housing, leading to a model of the form:

$$p_{i,t} = \beta_0 + \beta_1 p_{i,t-k} + \beta_2 y_{i,t-k} + \beta_3 u_{i,t-k} + \beta_4 c_{i,t-k} + \beta_5 r_{t-k}$$

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$$+ \beta_6 s_{t-k} + \beta_7 oil_{t-k} + \varepsilon_{i,t} \tag{1}$$

where p is the natural logarithm of real house prices in county i at time t. Housing prices are determined from the demand side by the purchasing parity expressed with the natural logarithm of real aggregate income y, the opportunity cost of capital measured by the real mortgage rate r, and the term spread s of the 10-year and the 3-month government securities as an indicator of market's expectation over future economic conditions. The use of the term spread and mortgage rates captures the monetary policy transmission channel of Koeniger et al. (2022). From the supply side, we consider the natural logarithm of the number of housing units u in each MSA that reflects housing supply, the natural logarithm of real oil prices[3] oil, and the log of construction cost c as cost determinants in housing production that can potentially affect supply.

The inclusion of the lagged value of house prices p on the right-hand side is motivated by potential feedbacks in housing demand that can cause momentum effects (Beracha & Skiba, 2011; Case & Shiller, 1989). Oil prices, interest rates, and the term spread are constant across all MSAs, while the construction cost is constant across counties of the same MSA. Thus, only income, housing units, and house prices vary over all cross-sections of our panel.

Equation (1) is extended by allowing county fixed effects and heterogeneity of slope coefficients across counties in (2), given that the relevant literature suggests that the strength of the momentum effect β_1 (Capozza et al., 2004), demand elasticities $\beta_2, \beta_5, \beta_6$ (Saiz, 2010), and supply elasticities $\beta_3, \beta_4, \beta_7$ vary across regions (Paciorek, 2013).

$$\begin{aligned} p_{i,t} = \beta_{0i} &+ \beta_{1i} p_{i,t-k} + \beta_{2i} y_{i,t-k} + \beta_{3i} u_{i,t-k} + \beta_{4i} c_{i,t-k} + \beta_{5i} r_{t-k} \\ &+ \beta_{6i} s_{t-k} + \beta_{7i} oil_{t-k} + \varepsilon_{i,t} \end{aligned} \tag{2}$$

Naturally, both (1) and (2) refer to long-term relationships that could be significantly different from short-term dynamics, since market frictions, construction lags, transaction costs, inefficient market information, and liquidity constraints do not allow for instantaneous supply and demand reactions (Glaeser et al., 2014). To account for short-term adjustments, we represent house price dynamics in an error-correction form (Harter-Dreiman, 2004, Oikarinen et al., 2018), where current period price changes are determined by lagged price changes and lagged growth rates of market fundamentals, along with deviations of house price levels from their long-term equilibrium in (3):

$$\begin{aligned} \Delta p_{i,t} = \alpha_{0i} &+ \alpha_{1i} \Delta p_{i,t-k} + \alpha_{2i} \Delta y_{i,t-k} + a_{3i} \Delta u_{i,t-k} + a_{4i} \Delta c_{i,t-k} + a_{5i} \Delta r_{t-k} + a_{6i} \Delta s_{t-k} \\ &+ a_{7i} \Delta \mathrm{oil}_{t-k} + a_{8i} (p - p*)_{i,t-1} + \varepsilon_{i,t}. \end{aligned} \tag{3}$$

The inclusion of the term $\Delta p_{i,t-k}$ reflects potential feedback in housing demand; high prices growth induces greater expectations and thus increases supply dynamics and ultimately house price growth rates. The error correction term $a_{8i}(p - p*)_{i,t-1}$ captures the delayed adjustment of house prices toward the long-term equilibrium price level, with parameter a_{8i} representing the adjustment speed, that is expected to be negative and between minus one and zero and stationary residual term $\varepsilon_{i,t}$. The term $p*$ represents the long-term equilibrium price of (2).

[3] Despite that oil prices in our sample exhibit a smaller variation in comparison with previous decades, we choose to include oil prices as a regressor as they are a key determinant of production cost and thus house prices.

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2.2 | Panel SVR

2.2.1 | Support vector regression

The SVR is a direct extension of the Support Vector Machines algorithm (Cortes and Vapnik, 1995). The method proposed by Vapnik (1995) originates from the field of statistical learning and its basic idea is to find a loss function that has at most a predetermined deviation (error) ε from the actual observations, instead of minimizing squared error. In other words, while constructing the optimal forecasting model, errors that do not violate a predefined threshold ε do not play any role in defining the model's parameters and are not considered. Only errors higher than ε are penalized. Thus, the SVR model can be considered as a "relaxed" regression model of higher flexibility, where the goal of the algorithm is not to find a strict line, but rather a more flexible error tolerance band. The vectors (data points) outside the error tolerance band, given ε, are identified through a minimization procedure and are called Support Vectors (SVs). The aforementioned estimator is typically referred to as the "e-insensitive" estimator or the "e-insensitive" loss function compared with the squared loss or Huber loss function used in the econometrics literature.

For a training dataset $D = [(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots .(\mathbf{x}_\mathrm{n}, y_n)]$, $\mathbf{x}_\mathrm{i} \in \mathbb{R}^n, y_i \in \mathbb{R}, i = 1, 2, \ldots .n$, where $\mathbf{x}_\mathrm{i}$ is a vector of independent variables and y_i is the dependent one, the linear SVR function takes the form $y = f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$. Then, the minimization problem is:

$$\min\left(\frac{1}{2}\mathbf{w}^2 + C\sum_{\iota=1}^{n}\left(\zeta_i + \zeta_i^*\right)\right) \tag{4}$$

$$\text{subject to}\begin{cases} y_i - (\mathbf{w}\mathbf{x}_\mathrm{i} + b) \le \varepsilon + \zeta_i \\ (\mathbf{w}\mathbf{x}_\mathrm{i} + b) - y_i \le \varepsilon + \zeta_i^* \\ \zeta_i,\ \zeta_i^* \ge 0 \end{cases}$$

where ε defines the width of the tolerance band, and ζ_i, ζ_i^* are slack variables controlled through a penalty parameter C. All points inside the error tolerance band have $\zeta_i,\ \zeta_i^* = 0$. The e-insensitive loss function is formalized as:

$$L_\varepsilon\left(y_i, f\left(x_i, w\right)\right) := \begin{cases} 0 & \text{if } |y_i - wx_i - b| < \varepsilon \\ |y_i - wx_i - b| - \varepsilon & \text{otherwise} \end{cases} \tag{5}$$

System (4) describes a convex quadratic optimization problem with linear constraints, ideally with a unique solution. The first part of the objective function controls the generalization ability of the regression, by imposing the "flatness" of the model controlled through the Euclidean norm $\mathbf{w}$, placing a penalty on model complexity. In other words, the first part of the objective problem imposes different weights on the variables (features) resulting in a more parsimonious model, acting as a regularization term. The second part of the objective function controls the fit of the regression to the training data (by increasing C, we penalize with a bigger weight any point outside the tolerance band, with $\zeta_i \ge 0$ or $\zeta_i^* \ge 0$) reducing the effect of outliers on the optimization function. The key element of the SVR concept is to find the balance between the two parts

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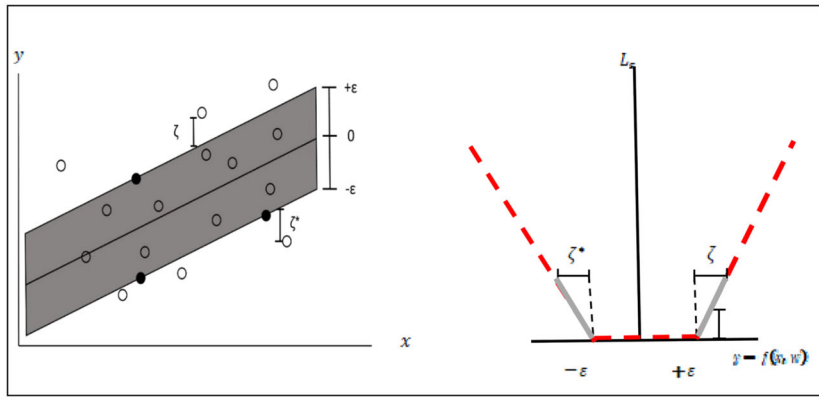

FIGURE 1 The tolerance band defined by ε. The boundaries of the error tolerance band are defined by the support vectors (SVs) denoted with the black filled circles (left plot). Forecasted values greater than $|\varepsilon|$ get a penalty equal to ζ that depends on their distance from the tolerance band (right plot). [Color figure can be viewed at wileyonlinelibrary.com]

of the objective function.[4] In Figure 1, we depict the notion of the error tolerance band and the e-insensitive loss function based on parameter ε.

Using the Lagrange multipliers in System (4) the optimization problem becomes

$$\mathcal{L} := \frac{1}{2}\sum_{i,j=1}^{N}\left(a_i - a_i^*\right)\left(a_i - a_i^*\right)\mathbf{x}_i^T\mathbf{x} - \sum_{i=1}^{N} y_i\left(a_i - a_i^*\right) + \varepsilon\sum_{i=1}^{N}\left(a_i - a_i^*\right) \tag{6}$$

given that $\sum_{i=1}^{N} y_i(a_i - a_i^*) = 0$ and $a_i, a_i^* \in [0, C]$

The solution is given by:

$$\mathbf{w} = \sum_{i=1}^{n}\left(a_i - a_i^*\right)\mathbf{x}_\mathrm{I} \tag{7}$$

and

$$y = \sum_{i=1}^{n}\left(a_i - a_i^*\right)\mathbf{x}_\mathrm{i}^\mathrm{T}\mathbf{x} + b \tag{8}$$

with the Lagrange multipliers α_i, $a_i^* = 0$ for all non-SVs. Thus, the SVR model is defined solely by its SVs.

One of the main advantages of SVR in comparison with other ML techniques (like for example an ANN trained with stochastic gradient descend or a RF that is a bootstrapped average of a large number of decision trees) is that it yields a convex minimization problem with a

[4] At the Supplementary File of the article, we provide an empirical exercise where we constrain our model to have C or ε equal zero. The empirical findings suggest that both the observation weighting by C and the error tolerance band contribute similarly to the forecasting performance of the SVR model.

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unique global minimum, avoiding local minima. The researcher has to determine exogenously the value of parameters C and ε in a way that maximizes the generalization ability of the model to other populations. In doing so, in the ML practice, one may find that the dataset is typically split in a train, validation, and test dataset. Selecting arbitrarily some staring values for the exogenously determined parameters, the values of all endogenous parameters are determined based on the train dataset through gradient descent and the forecasting performance is tested against the validation dataset. The values of the exogenous parameters are revised and the procedure is repeated until the minimum value of the loss function (5) at the validation dataset is achieved. Once the optimum forecasting performance is reached, all parameters are fixed and model's generalization ability is measured in the unknown test sample, kept aside during training.

However, this approach has the inherent disadvantage of depending heavily on the characteristics of the validation dataset. If the train and the validation datasets share common characteristics, the model will overfit[5] leading to poor generalization ability on the test dataset. To overcome this drawback, a cross-validation technique is employed during training, which ensures that the trained model does not suffer from overfitting and is expected to generalize well.[6] Naturally, cross-validation does not guarantee that any given model will adhere perfectly to any phenomenon, but ensures that it will not overfit due to researcher's assumptions during the training phase. Moreover, cross-validation potentially compensates for the existence of a unit root process and heteroskedasticity of the data that would make all results sample-specific, since it provides an optimum selection of the exogenous parameters.

The functional form of the model in Equation (6) is an affine one, while underlying data generating processes in real-life phenomena are rarely linear. Thus, deriving linear models to describe them often fails to grasp the true data generating process. To address this issue, SVR is typically coupled with kernel functions. The basic intuition of a kernel function is to map a nonlinear dataset in a richer dimensional space where a linear solution is feasible. Then, the linear solution is projected to the initial dimensional space in a nonlinear form. For illustration purposes, we provide a simple example of a kernel projection in the manuscript's Supplementary File.

In our empirical estimations, we evaluate two alternative kernels: the linear kernel and the nonlinear Radial Basis Function (RBF) kernel. Each kernel representation is:

$$\text{Linear } K_1(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^{\mathrm{T}} \mathbf{x}_2 \tag{9}$$

$$\text{RBF } K_2(\mathbf{x}_1, \mathbf{x}_2) = e^{-\gamma \mathbf{x}_1 - \mathbf{x}_2^2} \tag{10}$$

with factor γ representing a kernel parameter, determined exogenously during the cross-validation training scheme. The linear kernel does not project the data to a higher dimensional space, while the RBF kernel can be Taylor-expanded to an infinite-dimensional space. Fortunately, we do not need the actual mapping of the projected dataset to compute the SVR model, but only the dot products (values) from the kernel functions to compute our parameters. With the introduction of

[5] Adhering to the data and not the actual phenomenon. In other words, the model learns to forecast the specific data, but fails to discern the underlying data generating process.

[6] In the Supplementary File, we provide a detailed exposition of the cross-validation procedure used in training.

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a kernel function $K(\mathbf{x_i}, \mathbf{x_j}) = \varphi(\mathbf{x_i}), \varphi(\mathbf{x_j})$, the optimization problem (6) becomes

$$\min_{\mathrm{a},\mathrm{a}^*} := \frac{1}{2}\left[(\mathrm{a}^*)^T, \mathrm{a}^T\right]\begin{bmatrix}\mathrm{K} & -\mathrm{K}\\ -\mathrm{K} & \mathrm{K}\end{bmatrix}\begin{bmatrix}\mathrm{a}^*\\ \mathrm{a}\end{bmatrix} + \left[\varepsilon e^T - y^T,\ \varepsilon e^T + y^T\right]\begin{bmatrix}\mathrm{a}^*\\ \mathrm{a}\end{bmatrix} \tag{11}$$

given that $Z^T\left[\begin{smallmatrix}\mathrm{a}^*\\ \mathrm{a}\end{smallmatrix}\right] = 0$ and $a_i, a_i^* \in [0, C] \forall\, i = 1, 2, , \ldots, N$

$$\text{where } Z = \left[\underbrace{1, \ldots, 1,}_{N}\ \underbrace{-1, \ldots ., -1}_{N}\right]^T$$

The solution can be written as:

$$\mathbf{w} = \sum_{i=1}^{n}\left(a_i - a_i^*\right)\Phi(\mathbf{x_i}) \tag{12}$$

and

$$y = \sum_{i=1}^{n}\left(a_i - a_i^*\right)K\left(\mathbf{x}_i,\ \mathbf{x}\right) + \mathrm{b} \tag{13}$$

A drawback of using kernel methods is that we cannot obtain coefficients $\boldsymbol{w}$ for both kernels. Extraction of an analytical expression for the SVR model is available only for the linear kernel, where the model is projected to its original dimensional space and the coefficients a_i of the dual form corresponded to the independent variables.[7] In the nonlinear case (i.e., RBF kernel), the parameters of the mapping function cannot be matched to the independent variables, since it practically expands to a higher dimensional space adding more coefficients. This lack of natural interpretation of model's coefficients is not new to the ML literature (i.e., the weights in the synapses of neurons in an ANN are simply the values of derivatives lacking natural interpretation), that often leads to treating ML methods as "black boxes." Afterall, regression coefficients from ML methods are rarely consistent, as these methods are not built with a focus on statistical inference, but to generalize well (Athey and Imbens, 2019; Mullainathan and Spiess, 2017). In the same vein, standard errors cannot be derived asymptotically given that the exact distribution of the coefficients is unknown. To overcome this drawback, only in the linear kernel case where coefficients are tractable, we follow the stationary bootstrap procedure[8] of Politis and Romano (1994) to simulate the 16th and 84th percentiles of the empirical distribution.

[7] To be accurate, a number of studies attempt to approximate heuristically the exact value of SVR coefficients based on nonlinear kernels, but with limited success. See for instance Brown et al. (2012).

[8] The stationary bootstrap is merely a block bootstrap procedure that maintains the temporal ordering of the data to maintain the (possible) autocorrelation between observations.

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2.2.2 | Panel SVR

Although SVR solves several issues of the typical regression model, in the case of panel data, it fails to account for the cross-sectional dependence among cross-sections. To introduce this issue to the SVR model, we embed cross-correlations directly to the kernel matrix, while introducing heterogenous coefficients across cross-sections. Suppose that we have panel observations collected from N entities (counties in price forecasting or bubble detection) at T occasions/periods (months) with K variables (features). The expanded variable matrix x_{it} takes the form $NT \times K$, where $\mathbf{y}$ is an $\mathrm{NT} \times 1$ vector of $\mathbf{y_i} = [y_{i1}, y_{i2}, \dots, y_{it}]^T$ column vectors of size $T \times 1$. A typical heterogenous time-varying panel model would be written in the form

$$y_{it} = d_i + x_{it}^T \beta_{it} + u_{it} \tag{14}$$

where d_i are individual specific effects, β_{it} are K-dimensional vectors of time-varying heterogenous coefficients over i and model errors u_{it} are stationary over time t but may be cross-sectionally dependent. In the econometrics literature, the most prominent approach to deal with this category of models is to introduce a sieve approximation for the time-varying coefficient functions (Su et al., 2019) or kernel regressions in panel data (Chen, 2019). In the ML context, we cannot observe these heterogenous coefficients directly, especially in the nonlinear RBF kernel case. So, building on the work of Du et al. (2015), we need to establish cross-sectional heterogeneity directly to the kernel matrix prior to projection.

We extend model (11) by introducing the $\mathrm{NT} \times 1$ time-varying parameter vector $\mathbf{c_i} = [1, c_{i1}, c_{i2}, \dots, c_{iT-1}]^T$ of column vectors of size $T \times 1$. So, for each entity i we formulate our dataset as:

$$x_{i1} + c_{i1} x_{i2} + \cdots + c_{iT-1} x_{it} = x_{it}^T \mathbf{c_i} \tag{15}$$

$$y_{i1} + c_{i1} y_{i2} + \cdots + c_{iT-1} y_{it} = \mathbf{y_{it}^T c_i} \tag{16}$$

where the unknown parameter vector $\mathbf{c_i}$ takes into account observational dependence within sections (entities) and evolves with time. The introduction of parameter c both to x_{it} and y_{it} warranties that the SVR model will observe heterogenous cross-sectional dependencies. Thus, the Lagrangian function transforms to:

$$\begin{aligned} \mathcal{L} := \frac{1}{2} w^T w + C \sum_{i=1}^{N} \left(\zeta_i + \zeta_i^*\right) - \sum_{i=1}^{N} \left(\eta_i \zeta_i + \eta_i^* \zeta_i^*\right) \\ - \sum_{i=1}^{N} a_i \left(\varepsilon + \zeta_i - y_i^T c_i + w, x_i^T c_i + b\right) - \sum_{i=1}^{N} \alpha_i^* \left(\varepsilon + \zeta_i - y_i^T c_i + w, x_i^T c_i - b\right) \end{aligned} \tag{17}$$

under the constraints of the Lagrange multipliers $a_i, \alpha_i^{,*}, \ \eta_i, \eta_i^*$

$$\frac{\partial \mathcal{L}}{\partial c} = \sum_{i=1}^{n} \left(a_i - a_i^*\right) = 0 \tag{18}$$

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$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^{n} \left(a_i - a_i^*\right) \mathbf{x_{it} c_i} = 0 \tag{19}$$

$$\frac{\partial \mathcal{L}}{\partial \zeta_i} = C - a_i - \eta_i = 0 \tag{20}$$

$$\frac{\partial \mathcal{L}}{\partial \zeta_i^*} = C - a_i^* - \eta_i^* = 0 \tag{21}$$

Thus, the dual space quadratic problem can be written as:

$$\max := -\frac{1}{2} \sum_{i,j=1}^{N} \left(a_i - a_i^*\right) \left(a_j - a_j^*\right) \mathbf{c_i^T x_{it} x_j^T c_i} + \sum_{i=1}^{n} \mathbf{y_i^T c_i} \left(a_i - a_i^*\right) - \varepsilon \sum_{i=1}^{N} \left(a_i - a_i^*\right) \tag{20}$$

given $\sum_{i=1}^{n} (a_i - a_i^*) = 0$ and $a_i, a_i^* \epsilon [0, C]$

In matrix form, the optimization process yields:

$$\min_{\mathbf{a_m}} := \frac{1}{2} \mathbf{a}_m \begin{bmatrix} \mathbf{K_m} & -\mathbf{K_m} \\ -\mathbf{K_m} & \mathbf{K_m} \end{bmatrix} \mathbf{a_m} + \mathbf{y_m^T a_m} \tag{22}$$

with

$$\mathbf{a_m} = \left[a^{*^T} c_1 a^{*^T} \cdots c_{T-1} a^{*^T} a^T c_1 a^T \cdots c_{T-1} a^T\right]^T$$

$$\mathbf{y_m} = \left[\varepsilon e^T - y_1^T - y_2^T \cdots - y_t^T \varepsilon e^T + y_1^T y_2^T \cdots y_t^T\right]^T$$

$$G = \begin{bmatrix} K(\mathbf{x}_1, \mathbf{x}_1) & \cdots & K(\mathbf{x}_1, \mathbf{x_t}) \\ \vdots & \ddots & \vdots \\ K(\mathbf{x_T}, \mathbf{x}_1) & \cdots & K(\mathbf{x_T}, \mathbf{x_t}) \end{bmatrix}$$

$$a_i, a_i^* \epsilon [0, C]$$

$$\sum_{i=1}^{n} \left(a_i - a_i^*\right) = 0$$

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$$\mathbf{z}^T \begin{bmatrix} \mathbf{a}^* \\ \mathbf{a} \end{bmatrix} = 0$$

with no constraint imposed on parameter **c**. So, after determining a_m using gradient descent, we can estimate **c** that depicts the responses of each cross-section per time period.

The use of the kernel projection, the intolerance band, the nonlinear weighing of observations based on the kernel function and the selection of exogenous parameters C, γ, and ε based on a cross-validation scheme suggests that the SVR model is expected to outperform other typical statistical approaches in out-of-sample forecasting. Even so, there is no guarantee that more complex functional forms and training methods outperform simpler ones. Thus, we evaluate a plethora of popular econometric and ML methods to highlight the necessity of considering the panel structure of the data, regardless of the selected approach.

Another point of interest is the use of the error-correction functional form for all models in our forecasting exercises. The interest in endogenous validity is limited to the ML literature as the exact distribution of the estimators is rarely available and thus statistical inference is not relevant, with the interest in the external validity and the generalization ability of the model, relying heavily on the cross-validation training scheme. In contrast, endogenous validity that warrants asymptotic consistency of the econometric estimators is of keen interest to the econometric literature. Therefore, to provide equal opportunities to both families of models and not bias toward the ML one, we use the error-correction functional form throughout the article.

2.3 | The data

Starting from the house prices forecasting module, we compile a monthly dataset of house prices at the county level for the 56 most populated MSAs (over one million residents) from January 2000 to May 2022, from Zillow (2022). Our sample includes 431 counties, containing almost half of the U.S. population and over two-thirds of its urban population.[9] The sample includes data for all houses with 1, 2, 3, 4, and 5+ bedrooms and contains almost a full cycle of house prices, including the significant increase in house prices during the 2000s, the great drop during the global financial crisis of 2008, the sluggish recovery during the 2010s, and the great increase in real estate prices during the COVID-19 pandemic.

The mortgage interest rate is retrieved from the Federal Housing Finance Agency, while West-Texas intermediate oil prices and the term spread of the 10-year with the 3-month Treasury bills are from the database of the Federal Reserve of St. Louis. Annual construction costs are from the RS Means database (RSMeans, 2020) and are interpolated to monthly values according to changes in the shelter only component of the urban Consumer Price Index of the Bureau of Economic Analysis (BEA).

The personal income per county series of the BEA is provided annually, and we interpolate monthly values based on quarterly changes in the state (where available) or national (Gross Domestic Product) GDP from BEA, with spline interpolation from quarterly to monthly frequency. House prices and construction costs are deflated by the national urban (Consumer Price Index) CPI less shelter, while aggregate income and interest rates are deflated by the national

[9] More details can be found in the Supplementary File.

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urban CPI for all items. Finally, housing units per county are from U.S. Census and are interpolated from annual to monthly values using spline interpolation. Table 1 provides summary statistics for all variables in logarithmic first differences, at the MSA level, since it requires extensive space to provide distribution moment for all 431 counties.[10]

As depicted in Table 1, the selected sample contains significant heterogeneity, since six MSAs contain only one county (Fresno, Las Vegas-Henderson-Paradise, San Diego-Carlsbad, St. Louis, Tucson and Urban Honolulu), whereas other MSAs exhibit a much higher number of counties as Atlanta-Sandy Springs-Roswell with 29, 23 in New York-Newark-Jersey City, and 20 counties in Washington-Arlington-Alexandria, respectively. Regarding the house prices growth rate, San Diego-Carlsbad has a mean growth rate of 0.334 with a standard deviation of 1.105, Riverside-San Bernardino-Ontario a growth rate of 0.320 with a standard deviation of 1.373 and Miami-Fort Lauderdale-West Palm Beach 0.293 with a standard deviation of 0.804, while St. Louis a negative growth rate of −0.078 (standard deviation 0.661) and Salt Lake City −0.077 (standard deviation 1.135). In contrast, New Orleans-Metairie has a growth rate of 0.004 (standard deviation0.689), Chicago-Naperville-Elgin and Cleveland-Elyria have a growth rate of 0.011 (standard deviations 0.711 and 0.652, respectively), followed by Memphis with a growth rate of 0.014 (standard deviation 0.658). The null hypothesis of equal means across counties is rejected for 22 out of the 56 MSAs, supporting the assumption of spatial heterogeneity.

The spatial heterogeneity element is also apparent in the average income variable, where Austin-Round Rock, Phoenix-Mesa-Scottsdale, and San Antonio-New Braunfels exhibit the highest income growth rates (0.482, 0.550, and 0.301, respectively) and Buffalo-Cheektowaga-Niagara Falls, Detroit-Warren-Dearborn, Hartford-West Hartford-East Hartford, and Milwaukee-Waukesha-West Allis the lowest growth rates (0.089, 0.049, 0.082, and 0.097, respectively). For 37 out of the 56 MSAs, the null hypothesis of equal means across counties is rejected. The heterogeneity is even larger in the growth rates of housing units, with Austin-Round Rock, Dallas-Fort Worth-Arlington, Orlando-Kissimmee-Sanford, and Phoenix-Mesa-Scottsdale exhibiting a high growth rate (0.302, 0.228, 0.201, and 0.212 respectively) and other MSAs having almost no growth in the number of housing units (Buffalo-Cheektowaga-Niagara Falls: 0.022, New Orleans-Metairie: 0.009, Pittsburgh: 0.012, and Rochester: 0.035). The null hypothesis of equal means across counties is rejected for 51 out of the 56 MSAs. In contrast, the construction cost variable seems to be similar across MSAs, with a growth rate around 0.350 for all regions. The statistical test of heterogeneity across counties is irrelevant, as construction cost is available at the MSA level.

The second forecasting simulation evaluates the ability of all models to identify bubble formation in the real estate market. In doing so, we compile a dataset of house rent prices from Zillow (2022) and the Bureau of Labor Statistics (BLS). More specifically, Zillow price rents are available for the period January 2014–May 2022, while BLS publishes monthly indices of rent prices that go back to 1986. Using micro-data from the 2000 and 2020 Decennial Census of Housing we convert our county house prices to price–rent ratios for the period 2000–2014, following the procedure described in Campbell et al. (2009). The price–rent ratios are used as the benchmark (ground truth) for the existence of a bubble.

Finally, to access the economic significance of the research question and provide a meaningful contribution in terms of practical implication to practitioners, we compile a dataset of 184 residential REITs from the National Association of Real Estate Investment Trust (2022), available for 40 U.S. states, at the state level. All REITs in our sample are actively traded in the stock market. We

[10] The correlation matrix for all variables, stacked by county, is reported in the Supplementary File. A number of significant correlations arise, but since they do not alter the conclusions, we do not report them in the main body of the article.

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TABLE 1 Summary statistics.

Metropolitan statistical area (MSA)	State (1)	Number of counties (2)	Total observations (3)	Real house price growth (Δp) (4)	Real aggregate income growth (Δy) (5)	Housing units growth (Δu) (6)	Real construction costs growth (Δc) (7)
Atlanta-Sandy Springs-Roswell	GA	29	7801	0.109 (0.950)***	0.186 (0.473)***	0.151 (0.191)***	0.357 (0.354)
Austin-Round Rock	TX	5	1345	0.162 (0.806)**	0.317 (0.482)***	0.302 (0.321)***	0.358 (0.339)
Baltimore-Columbia-Towson	MD	7	1883	0.132 (0.821)	0.138 (0.416)	0.085 (0.083)***	0.363 (0.324)
Birmingham-Hoover	AL	6	1614	−0.024 (0.599)***	0.156 (0.444)***	0.070 (0.141)***	0.36 (0.367)
Boston-Cambridge-Newton	MA	7	1883	0.200 (0.808)	0.151 (0.448)	0.073 (0.067)***	0.337 (0.298)
Buffalo-Cheektowaga-Niagara Falls	NY	2	538	0.162 (0.600)	0.089 (0.412)	0.022 (0.039)***	0.364 (0.305)
Charlotte-Concord-Gastonia	NC	11	2959	0.131 (0.738)	0.211 (0.522)***	0.137 (0.228)***	0.402 (0.398)
Chicago-Naperville-Elgin	IL	14	3766	0.011 (0.711)***	0.136 (0.497)***	0.095 (0.147)***	0.373 (0.387)
Cincinnati	OH	16	4304	0.032 (0.625)*	0.128 (0.566)***	0.054 (0.137)***	0.337 (0.313)
Cleveland-Elyria	OH	5	1345	0.011 (0.652)	0.094 (0.434)***	0.060 (0.063)***	0.323 (0.294)
Columbus	OH	10	2690	0.076 (0.853)	0.177 (0.438)***	0.100 (0.148)***	0.381 (0.436)
Dallas-Fort Worth-Arlington	TX	11	2959	0.119 (0.777)**	0.268 (0.474)***	0.228 (0.244)***	0.358 (0.333)
Denver-Aurora-Lakewood	CO	10	2690	0.182 (0.736)***	0.172 (0.588)***	0.154 (0.180)***	0.341 (0.331)
Detroit-Warren-Dearborn	MI	6	1614	−0.020 (1.099)	0.049 (0.460)**	0.042 (0.082)***	0.331 (0.400)
Fresno	CA	1	269	0.245 (1.269)	0.205 (0.443)	0.091 (0.047)	0.367 (0.299)
Grand Rapids-Wyoming	MI	4	1076	0.092 (0.774)	0.117 (0.461)	0.058 (0.094)***	0.367 (0.315)
Hartford-West Hartford-East Hartford	CT	3	807	0.069 (0.693)	0.082 (0.419)	0.051 (0.052)***	0.356 (0.302)
Houston-The Woodlands-Sugar Land	TX	9	2421	0.136 (0.599)***	0.278 (0.509)***	0.207 (0.211)***	0.342 (0.595)
Indianapolis-Carmel-Anderson	IN	11	2959	0.024 (0.619)***	0.138 (0.444)***	0.112 (0.132)***	0.343 (0.433)
Jacksonville	FL	5	1345	0.245 (0.959)	0.238 (0.482)***	0.181 (0.156)***	0.391 (0.475)

(Continues)

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TABLE 1 (Continued)

Metropolitan statistical area (MSA)	State (1)	Number of counties (2)	Total observations (3)	Real house price growth (Δp) (4)	Real aggregate income growth (Δy) (5)	Housing units growth (Δu) (6)	Real construction costs growth (Δc) (7)
Kansas City	MO	14	3766	0.080 (0.688)***	0.122 (0.525)***	0.052 (0.192)***	0.353 (0.305)
Las Vegas-Henderson-Paradise	NV	1	269	0.176 (1.501)	0.220 (0.520)	0.185 (0.155)	0.355 (0.321)
Los Angeles-Long Beach-Anaheim	CA	2	538	0.328 (1.069)	0.150 (0.457)	0.054 (0.036)***	0.371 (0.328)
Louisville/Jefferson County	KY	10	2690	0.018 (0.630)***	0.165 (0.451)***	0.097 (0.141)***	0.344 (0.349)
Memphis	TN	8	2152	0.014 (0.658)***	0.154 (0.535)***	0.081 (0.192)***	0.366 (0.346)
Miami-Fort Lauderdale-West Palm Beach	FL	3	807	0.281 (1.186)	0.173 (0.535)	0.094 (0.088)***	0.364 (0.400)
Milwaukee-Waukesha-West Allis	WI	4	1076	0.109 (0.780)	0.097 (0.427)	0.077 (0.068)***	0.347 (0.276)
Minneapolis-St. Paul-Bloomington	MN	15	4035	0.100 (0.865)***	0.182 (0.446)***	0.132 (0.131)***	0.334 (0.310)
Nashville-Davidson–Murfreesboro–Frank	TN	13	3497	0.176 (0.721)***	0.190 (0.466)***	0.144 (0.140)***	0.368 (0.337)
New Orleans-Metairie	LA	8	2152	0.004 (0.689)***	0.119 (0.750)***	0.009 (1.394)***	0.367 (0.359)
New York-Newark-Jersey City	NY	23	6187	0.158 (0.823)***	0.109 (0.453)***	0.057 (0.093)***	0.327 (0.24)
Oklahoma City	OK	7	1883	0.103 (0.630)	0.219 (0.466)***	0.133 (0.264)***	0.362 (0.362)
Orlando-Kissimmee-Sanford	FL	4	1076	0.216 (1.227)	0.253 (0.479)***	0.201 (0.208)***	0.372 (0.400)
Philadelphia-Camden-Wilmington	PA	11	2959	0.136 (0.780)	0.120 (0.450)**	0.066 (0.082)***	0.299 (0.370)
Phoenix-Mesa-Scottsdale	AZ	2	538	0.244 (1.423)	0.344 (0.550)***	0.212 (0.275)***	0.364 (0.358)
Pittsburgh	PA	7	1883	0.102 (0.594)	0.109 (0.440)*	0.012 (0.084)***	0.355 (0.303)
Portland-Vancouver-Hillsboro	OR	7	1883	0.237 (0.846)	0.188 (0.458)	0.105 (0.089)***	0.371 (0.337)
Providence-Warwick	RI	6	1614	0.231 (0.919)	0.114 (0.437)	0.048 (0.065)***	0.351 (0.323)

(Continues)

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TABLE 1 (Continued)

Metropolitan statistical area (MSA)	State (1)	Number of counties (2)	Total observations (3)	Real house price growth (Δp) (4)	Real aggregate income growth (Δy) (5)	Housing units growth (Δu) (6)	Real construction costs growth (Δc) (7)
Raleigh	NC	3	807	0.110 (0.739)***	0.236 (0.446)*	0.209 (0.131)***	0.394 (0.372)
Richmond	VA	14	3766	−0.013 (0.656)	0.168 (0.477)***	0.109 (0.136)***	0.357 (0.330)
Riverside-San Bernardino-Ontario	CA	2	538	0.320 (1.373)	0.222 (0.447)*	0.110 (0.108)***	0.353 (0.291)
Rochester	NY	6	1614	0.101 (0.646)	0.100 (0.431)	0.035 (0.139)***	0.353 (0.378)
Sacramento–Roseville–Arden-Arcade	CA	4	1076	0.267 (1.225)***	0.208 (0.431)*	0.125 (0.108)***	0.356 (0.266)
Salt Lake City	UT	2	538	−0.077 (1.135)***	0.277 (0.465)***	0.178 (0.103)***	0.364 (0.333)
San Antonio-New Braunfels	TX	8	2152	0.029 (0.656)	0.301 (0.477)***	0.199 (0.294)***	0.357 (0.333)
San Diego-Carlsbad	CA	1	269	0.334 (1.105)	0.157 (0.431)	0.065 (0.040)	0.370 (0.334)
San Francisco-Oakland-Hayward	CA	5	1345	0.277 (1.145)	0.184 (0.518)	0.055 (0.061)***	0.363 (0.278)
San Jose-Sunnyvale-Santa Clara	CA	2	538	0.225 (1.159)***	0.189 (0.563)	0.092 (0.087)***	0.364 (0.283)
Seattle-Tacoma-Bellevue	WA	3	807	0.281 (0.958)	0.214 (0.473)	0.114 (0.046)***	0.364 (0.308)
St. Louis	MO	15	4035	−0.078 (0.661)	0.117 (0.436)***	0.036 (0.263)***	0.340 (0.321)
Tampa-St. Petersburg-Clearwater	FL	4	1076	0.272 (1.148)	0.197 (0.464)***	0.119 (0.108)***	0.374 (0.420)
Tucson	AZ	1	269	0.177 (0.993)	0.175 (0.443)	0.098 (0.063)	0.356 (0.339)
Tulsa	OK	7	1883	0.090 (0.666)	0.158 (0.527)***	0.032 (0.207)***	0.363 (0.330)
Urban Honolulu	HI	1	269	0.293 (0.804)	0.139 (0.408)	0.081 (0.101)	0.334 (1.413)
Virginia Beach-Norfolk-Newport News	VA	16	4304	0.080 (0.786)***	0.146 (0.450)***	0.081 (0.188)***	0.380 (0.355)
Washington-Arlington-Alexandria	DC	20	5380	0.101 (0.937)***	0.200 (0.455)***	0.121 (0.152)***	0.352 (0.390)

Note: The sample period is February 2000–May 2022. Standard deviations in parentheses. To enhance readability all values are multiplied by 100. *, **, and *** denote differences between county means with statistical significance at the 10, 5, and 1% level, respectively. Equality of means is tested with a Welch F-test on the null hypothesis of equal means across counties.

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compile their monthly stock returns from February 2000 to May 2022 from DataStream (2022). All company names are available in the Supplementary File. In Table 2, we report the descriptive statistics of the REIT dataset.

In each state, we construct an equally weighted monthly portfolio of REIT stocks that are active in the selected state. Our approach directly follows Jacobs et al. (2014), who show that equally weighted trading portfolios outperform value-weighted ones in terms of performance and especially in out-of-sample forecasting. Given that many REITs are active in more than one state, they are included in more than one state portfolio. Naturally, not all REITs are publicly traded for the entire 22 years included in our sample. As we observe from the last column of Table 2, our sample includes on average 21% (Georgia) to 52% (Delaware) of the REITs that are actively traded in May 2022 (end of our sample) in the stock market. Thus, our sample includes a representative sample of the entire population for each state. The average return over the entire period of our sample is 2% in Georgia, 7.4% in Louisiana, and 3.1% in New Jersey, while it is 23.2% in Alabama, 26% in Mississippi, 28.6% in New Hampshire, 33% in Rhode Island, and 27.4% in Washington, suggesting significant heterogeneity.

3 | EMPIRICAL FINDINGS

To cater for a fair comparison between econometric and panel SVR models, we utilize estimators consistent with the existence of cross-sectional dependence, spatial heterogeneity, and nonstationarity (Holly et al, 2010). Table 3 provides a summary of the methods that we employ.

More specifically, we consider the MG (Pesaran & Smith, 1995) estimator that allows for cross-sectional heterogenous coefficients and nonstationary (but cointegrated) data, the augmented mean group (AMG) (Eberhardt & Teal, 2010) estimator that incorporates a common dynamic process regressor to control for cross-sectional correlation, the common correlated effects mean group (CCEMG) (Pesaran, 2006) estimator that controls for cross-sectional dependence in addition to the characteristics of MG, and the dynamic common correlated effects mean group (DCCEMG) (Chudik & Pesaran, 2015) that addresses endogeneity issues through the use of unobserved factors (these need not to be determined, but are represented through lagged means of cross-sections) rendering the estimator unbiased and consistent.

The AMG and the CCEMG estimators can be treated as different facets of the same approach, with the AMG model approaching the common dynamic process with a pooled OLS regression on sectional dummies, while the CCEMG treats common dynamic factors as nuisance parameters used solely for controlling cross-sectional dependence. Moreover, the CCEMG and the DCCEMG estimators also impose a regularization strategy through the imposition of different weights in the exogenous variables and the unobserved factors, in the vein of the normalization norm (first part) of the SVR model that levels the comparison field between the two methodologies. For comparison reasons, we also include two homogenous panel estimators; the conventional two-way fixed effects (TWFE) and the system GMM (Blundell & Bond, 1998) estimator, with the latter accounting for possible endogeneity of the TWFE estimator.

From the family of ML estimators, we evaluate the forecasting performance of the typical SVR model (that does not cater for the panel structure of the data), the RF estimator, an ANN model based on stochastic gradient descent training, the popular Least Absolute Shrinkage and Selection Operator (LASSO) estimator, and the XGBoost; an ensemble model based on boosting on multiple decision trees, unlike RF estimators that are based on bagging (bootstrapping) to create an ensemble of decision trees. All ML estimators' structure and exogenous parameters are

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TABLE 2 REIT descriptive statistics.

State	Number of REITS (1)	Average return (2)	Standard deviation (3)	Average number of active REITs (4)	Average percentage of active REITs (5)
Alabama	83	0.232	0.043	28	0.34
Arkansas	55	0.212	0.042	20	0.36
Arizona	30	0.095	0.047	8	0.25
California	144	0.205	0.048	53	0.37
Colorado	119	0.195	0.046	47	0.39
Connecticut	76	0.200	0.052	33	0.43
District of Columbia	30	0.128	0.061	15	0.50
Delaware	36	0.214	0.047	19	0.52
Florida	158	0.191	0.048	58	0.37
Georgia	30	0.020	0.046	6	0.21
Hawaii	29	0.150	0.044	12	0.42
Illinois	128	0.165	0.045	45	0.35
Indiana	91	0.128	0.046	35	0.38
Kansas	56	0.124	0.045	23	0.40
Kentucky	74	0.221	0.045	28	0.38
Louisiana	30	0.074	0.044	10	0.34
Massachusetts	104	0.165	0.051	42	0.41
Maryland	30	0.130	0.050	10	0.32
Michigan	90	0.181	0.048	32	0.36
Minnesota	84	0.135	0.044	32	0.38
Missouri	85	0.161	0.045	31	0.36
Mississippi	56	0.260	0.044	18	0.33
North Carolina	30	0.106	0.051	9	0.30
New Hampshire	43	0.286	0.054	21	0.49
New Jersey	30	0.031	0.046	8	0.28
Nevada	76	0.218	0.050	27	0.36
New York	114	0.188	0.047	51	0.44
Ohio	103	0.216	0.045	39	0.38
Oklahoma	30	0.139	0.041	8	0.27
Oregon	80	0.177	0.049	32	0.40
Pennsylvania	111	0.154	0.048	45	0.41
Rhode Island	30	0.330	0.049	10	0.32
South Carolina	101	0.180	0.044	38	0.37
Tennessee	102	0.212	0.050	40	0.39
Texas	147	0.191	0.047	54	0.37
Utah	64	0.158	0.045	22	0.35
Virginia	132	0.186	0.045	51	0.39
Washington	101	0.274	0.046	41	0.40
Wisconsin	66	0.207	0.043	23	0.35
West Virginia	30	0.128	0.037	10	0.33

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TABLE 3 Estimators considered.

Method/estimator	Characteristics	Source
Panel A: Econometric estimators		
Two-way fixed effect (TWFE)	Fixed effect panel regression model, controlling for time and cross-sectional individual effects to removing unit-specific time averages and changes that have the same effect on all units.	
System generalized moments estimator (sysGMM)	Control for endogeneity issues of independent variables including lagged values of the independent variables as instruments, assuming that the first differences of instrument variables are uncorrelated with the fixed effects.	Blundell and Bond (1998)
Mean group (MG)	Allows for spatial heterogeneity and nonstationary, but cointegrated, data	Pesaran and Smith (1995)
Augmented mean group (AMG)	Common dynamic process (sectional dummies) regressor to control for cross-sectional correlation	Eberhardt and Teal (2010)
Common correlated effects mean group (CCEMG)	Controls for cross-sectional dependence in addition to the characteristics of MG regressing to cross-sectional means in the sense of a factor regression model	Pesaran (2006)
Dynamic CCEMG (DCCEMG)	Addresses endogeneity issues in addition to CCEMG, using lagged cross-sectional means as unobserved factors	Chudik and Pesaran (2015)
Panel B: Machine learning estimators		
Least absolute shrinkage and selection operator (LASSO)	Extend OLS by adding a regularization term to shrink variable coefficients and improve fit	Tibshirani (1996)
Random forest estimator (RF)	Ensemble of a large number of Decision Trees (bootstrap—bagging) to avoid overfitting	Breiman (2001)
Artificial neural network (ANN)	A collection of sigmoidal functions organized in subsequent layers connected with adjustable weights	
eXtreme gradient boosting (XGBoost)	Ensemble of a large number of Decision Trees trained based on Boosting (weighing more of trees with larger error in an iterative training scheme)	Chen and Guestrin (2016)
Support vector regression (SVR)	A flexible version of OLS regression, based on an error tolerance band coupled with kernel projection to accommodate nonlinearities	Vapnik (1995)

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determined based on a cross-validation training scheme. In our forecasting exercise, we follow a rolling window approach to account for potential structural breaks in variables over the 22 years of our dataset. More specifically, we train our models to a window of 60 observations (5 years) and forecast h periods ahead. Afterward, the window is shifted one period ahead, dropping the first observation in each cross-section and including one new observation. The process is repeated iteratively until the end of all available observations. Thus, all reported accuracies correspond to average RMSEs over the forecasted windows.

A mandatory first step for our econometric estimators to be consistent is to test for the existence of unit roots, so that the error-correction term is meaningful. After the examination for the existence of unit roots, we test long-run equations for panel cointegration and remaining heterogeneity, to fulfil all necessary assumptions on data imposed by the aforementioned heterogenous methods. The notion of stationarity and cointegration is of little interest to the ML literature, since cross-validation training caters for nonstationarity, heteroskedasticity, and disrupts autocorrelation issues.

3.1 | Unit root and cointegration analysis

We test the null hypothesis of nonstationarity with the cross-sectional augmented IPS test (CIPS) of Pesaran (2007) that augments conventional augmented Dickey–Fuller regressions with cross-sectional averages to control for spatial heterogeneity, since our entire approach is based on the notion of spatial heterogeneity and cross-sectional dependency. Table 4 reports the unit root test statistics for the panel and the univariate unit root tests on the variables that are constant in cross-sections.

The ADF statistics' values suggest that the nonstationarity assumption cannot be rejected for oil prices and interest rate yield at levels, while mortgage rates are stationary. In contrast, all series are stationary in first differences. The panel CIPS test concludes that all variables are stationary in first differences. We repeat unit root tests at the MSA level to ensure that all variables are stationary at first differences, examining 50 panel datasets, while for the six MSAs with only one county, we use conventional augmented Dickey–Fuller regressions.

Results presented in Panel A of Table 5, show rejection of the nonstationarity hypothesis of house prices at 32 MSAs, 17 MSAs for income, and 12 MSAs for housing units at the 5% level of significance. Regarding growth rates (first differences reported in Panel B), the rejection of the null hypothesis is unanimous at the 1% level of significance. Construction costs are evaluated using univariate unit root tests, since they are constant across all counties of the same MSA, supporting the use of first differences.

Given that most variables are nonstationary in levels and stationary in first differences, we test for the existence of cointegration that allows for an error-correction structure of the long- and short-run dynamics of house prices. Forming our data as a panel dataset with counties as cross-sections, the Kao, the Pedroni, and the Westerlund panel cointegration tests reject the null hypothesis of no cointegration (Table 6). Thus, the error-correction formulation of our forecasting model can be used to account for the common stochastic trend of the independent variables.

Prior to reporting forecasting errors, it is interesting to examine house prices dynamics on the entire sample to evaluate the ability of the proposed methodology to describe the underlying house prices dynamics. In doing so, we estimate models' coefficients to appraise heterogeneous estimators, training all models in the entire sample and examining the stationarity and heterogeneity of the estimated residuals. In Table 7, we report the coefficients of the long-term model (model in

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TABLE 4 Unit Root test results for variables that are constant across MSAs.

Variable	Levels				First differences			
	Cross-section dependence (1)	Augmented Dickey–Fuller (2)	Phillips Perron (3)	Kwiatkowski–Phillips–Schmidt–Shin (4)	Cross-section dependence (5)	Augmented Dickey–Fuller (6)	Phillips Perron (7)	Kwiatkowski–Phillips–Schmidt-Shin (8)
Real oil prices	–	−2.171	−2.480	0.411***	–	−7.785***	−13.481***	0.065
Real mortgage rates	–	−5.590***	−6.206***	0.033	–	−12.674***	−6.941***	0.083
Real interest yield	–	−2.568	−2.328	0.145*	–	−6.520***	−15.610***	0.045
Real house prices	−2.047*	–	–	–	−4.489***	–	–	–
Real county income	−1.594	–	–	–	−5.107***	–	–	–
Real housing units	−1.358	–	–	–	−5.106***	–	–	–
Real construction costs	−1.693	–	–	–	−5.151***	–	–	–

Note: ***, **, and * denotes rejection of the null hypothesis at the 1, 5, and 10% level of significance, respectively. All tests include a trend and an intercept. The optimum lag order of the test is determined according to the Bayesian Information Criterion. The Kwiatkowski–Phillips–Schmidt–Shin test assumes as the null hypothesis that the series are stationary, while the augmented Dickey–Fuller, the Cross- Section Dependence, and the Phillips Perron test the opposite.

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TABLE 5 Unit root test results.

Metropolitan statistical area (MSA)	Panel A: Levels				Panel B: First differences			
	Real house prices (1)	Real county income (2)	Housing units (3)	Construction cost (4)	Real house prices (5)	Real county income (6)	Housing units (7)	Construction cost (8)
Atlanta-Sandy Springs-Roswell	−1.706*	−1.583	−0.966	0.139	−4.498***	−2.795***	−2.772***	−6.172***
Austin-Round Rock	−1.567	−1.504	−0.689	0.757	−3.888***	−2.237***	−2.345***	−8.850***
Baltimore-Columbia-Towson	−2.625***	−2.367***	−2.126***	−2.225**	−4.741***	−3.361***	−2.628***	−4.653***
Birmingham-Hoover	−2.251**	−1.919*	−1.445	1.760	−2.984***	−3.177***	−3.439***	−8.912***
Boston-Cambridge-Newton	−1.464	−2.307**	−1.465	−1.083	−5.200***	−3.258***	−2.514***	−5.937***
Buffalo-Cheektowaga-Niagara Falls	−2.614***	−1.592	0.390	0.494	−4.839***	−3.571***	−2.808***	−8.277***
Charlotte-Concord-Gastonia	−1.621	−1.100	−0.667	0.414	−4.523***	−3.452***	−3.297***	−6.659***
Chicago-Naperville-Elgin	−1.606	−1.918*	−1.246	−1.461	−3.809***	−3.002***	−3.154***	−4.931***
Cincinnati	−2.184**	−1.619	−0.688	0.958	−4.532***	−4.489***	−2.484***	−8.354***
Cleveland-Elyria	−1.600	−1.675*	−0.047	0.885	−5.538***	−3.364***	−3.192***	−6.023***
Columbus	−2.012**	−1.390	−0.727	0.604	−6.190***	−3.146***	−3.092***	−7.784***
Dallas-Fort Worth-Arlington	−2.491***	−1.371	−1.291	0.758	−5.362***	−2.517***	−2.731***	−5.933***
Denver-Aurora-Lakewood	−3.160***	−1.667*	−1.087	−0.064	−4.026***	−3.040***	−2.972***	−4.862***
Detroit-Warren-Dearborn	−2.649***	−1.820*	−0.063	−0.114	−4.331***	−2.822***	−2.923***	−5.767***
Fresno (adf)	−2.817***	−3.288***	−1.236	−1.741	−2.421***	−2.448***	−4.879***	−3.498***

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TABLE 5 (Continued)

Metropolitan statistical area (MSA)	Panel A: Levels				Panel B: First differences			
	Real house prices (1)	Real county income (2)	Housing units (3)	Construction cost (4)	Real house prices (5)	Real county income (6)	Housing units (7)	Construction cost (8)
Grand Rapids-Wyoming	−0.546	−1.339	1.682	0.376	−5.415***	−2.429***	−4.232***	−7.580***
Hartford-West Hartford-East Hartford	−2.618***	−2.414***	1.030	−1.693	−5.574***	−5.089***	−3.151***	−5.170***
Houston-The Woodlands-Sugar Land	−2.623***	−1.703*	−1.857*	−0.207	−4.421***	−2.676***	−2.735***	−7.865***
Indianapolis-Carmel-Anderson	−2.191**	−1.024	−0.411	1.821	−4.347***	−2.669***	−3.086***	−7.808***
Jacksonville	−1.447	−1.578	−0.371	−1.306	−3.715***	−2.592***	−2.797***	−3.729***
Kansas City	−2.655***	−1.886*	0.758	0.393	−3.736***	−3.343***	−2.553***	−7.778***
Las Vegas-Henderson-Paradise (adf)	−2.326**	−2.695***	−2.378***	−1.523	−2.474***	−3.847***	−3.479***	−3.042***
Los Angeles-Long Beach-Anaheim	−2.814***	−1.725	−3.026***	−1.858	−4.266***	−3.734***	−2.964***	−4.028***
Louisville/Jefferson County	−2.197**	−1.593	−0.094	0.236	−4.668***	−2.388***	−2.725***	−7.501***
Memphis	−1.920*	−1.223	−1.309	−1.561	−4.347***	−5.089***	−2.325***	−8.055***
Miami-Fort Lauderdale-West Palm Beach	−3.580***	−1.966**	−2.320**	−1.775	−3.014***	−3.458***	−2.352***	−2.867***
Milwaukee-Waukesha-West Allis	−2.840***	−1.699	−1.143	−0.569	−5.525***	−2.397***	−2.043***	−6.459***
Minneapolis-St. Paul-Bloomington	−3.060***	−2.124***	−0.583	−0.787	−4.520***	−3.114***	−3.173***	−4.682***

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TABLE 5 (Continued)

Metropolitan statistical area (MSA)	Panel A: Levels				Panel B: First differences			
	Real house prices (1)	Real county income (2)	Housing units (3)	Construction cost (4)	Real house prices (5)	Real county income (6)	Housing units (7)	Construction cost (8)
Nashville-Davidson–Murfreesboro–Frank	−2.111**	−1.085	−1.858	0.826	−4.619***	−2.891***	−2.815***	−8.575***
New Orleans-Metairie	−2.164**	−1.822*	−2.106**	−0.562	−4.230***	−2.497***	−2.802***	−8.060***
New York-Newark-Jersey City	−1.266	−1.590	−1.757	−2.578***	−4.291***	−3.232***	−2.340***	−5.332***
Oklahoma City	−1.759*	−1.851*	−0.109	−2.408***	−5.061***	−3.322***	−3.010***	−6.357***
Orlando-Kissimmee-Sanford	−2.539***	−1.780*	−1.106	−0.856	−4.742***	−3.999***	−3.128***	−3.342***
Philadelphia-Camden-Wilmington	−1.802*	−2.201**	−1.358	−1.584	−4.354***	−2.694***	−2.922***	−5.198***
Phoenix-Mesa-Scottsdale	−2.524***	−2.955***	−2.610***	−1.346	−4.547***	−1.845*	−3.133***	−3.028***
Pittsburgh	−2.097**	−1.118	−0.433	0.298	−4.824***	−3.162***	−3.042***	−8.541***
Portland-Vancouver-Hillsboro	−1.850*	−1.201	−1.583	−1.074	−4.375***	−3.088***	−2.906***	−4.703***
Providence-Warwick	−1.901*	−1.920*	−1.909*	−1.631	−4.606***	−3.093***	−2.771***	−4.514***
Raleigh	−2.827***	−1.023	−0.578	2.523***	−5.739***	−2.187***	−2.978***	−5.987***
Richmond	−2.203***	−1.290	−0.697	0.975	−2.719***	−3.329***	−2.689***	−8.611***
Riverside-San Bernardino-Ontario	−1.632	−3.698***	−2.610***	−1.804	−6.190***	−2.502***	−2.332***	−3.297***
Rochester	−1.367	−1.613	−2.064**	−0.758	−4.040***	−2.605***	−4.313***	−6.636***
Sacramento–Roseville–Arden-Arcade	−2.121***	−0.924	−3.486***	−1.654	−5.368***	−3.520***	−3.059***	−4.605***
Salt Lake City	−2.349***	−3.026***	−2.531***	−0.142	−1.711***	−2.695***	−3.405***	−5.303***

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TABLE 5 (Continued)

Metropolitan statistical area (MSA)	Panel A: Levels				Panel B: First differences			
	Real house prices (1)	Real county income (2)	Housing units (3)	Construction cost (4)	Real house prices (5)	Real county income (6)	Housing units (7)	Construction cost (8)
San Antonio-New Braunfels	−1.043	−1.578	−0.123	−0.208	−3.364***	−3.172***	−3.244***	−7.723***
San Diego-Carlsbad (adf)	−1.944*	−2.728***	−3.071***	−1.594	−3.149***	−2.303***	−7.246***	−4.261***
San Francisco-Oakland-Hayward	−0.648	−1.169	−0.074	−1.411	−4.986***	−3.603***	−2.175***	−4.730***
San Jose-Sunnyvale-Santa Clara	−0.532	−1.189	−1.949*	−1.185	−5.272***	−2.093***	−2.912***	−4.348***
Seattle-Tacoma-Bellevue	−1.650	−1.711*	−1.524	−0.804	−3.591***	−2.096***	−3.241***	−4.867***
St. Louis (adf)	−1.011	−1.374	−0.592	−0.574	−2.880***	−2.343***	−2.594***	−8.071***
Tampa-St. Petersburg-Clearwater	−2.231***	−3.819***	1.952	−0.638	−5.155***	−2.780***	−3.086***	−3.184***
Tucson (adf)	−1.661	−2.660***	−2.745***	−0.908	−2.890***	−2.384***	−7.070***	−3.893***
Tulsa	−1.281	−2.104**	0.268	0.017	−5.628***	−2.662***	−2.858***	−7.973***
Urban Honolulu (adf)	−2.082***	−2.794***	0.321	−1.816	−3.679***	−2.676***	−6.573***	−5.805***
Virginia Beach-Norfolk-Newport News	−2.376***	−1.496	−0.638	0.839	−3.672***	−2.495***	−3.178***	−8.063***
Washington-Arlington-Alexandria	−3.011***	−2.082**	−0.766	−2.129**	−3.642***	−2.878***	−2.547***	−5.252***

Note: ***, **, and * denotes rejection of the null hypothesis about nonstationarity across panels controlling for heterogenous cross-sectional dependence (Pesaran, 2007) at the 1, 5, and 10% level of significance, respectively. adf denotes the use of the typical augmented Dickey–Fuller test for panels with one cross-section. All tests include a trend and an intercept. The optimum lag order of the test is determined according to the Bayesian Information Criterion. Construction costs are tested with the adf test, since costs are stable across counties of the same MSA.

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TABLE 6 Cointegration test results.

	Modified Dickey–Fuller (1)	Dickey–Fuller (2)	Augmented Dickey–Fuller (3)	Unadjusted modified Dickey–Fuller (4)	Unadjusted Dickey–Fuller (5)	Modified Phillips—Perron (6)	Phillips—Perron (7)	Variance ratio (8)
Kao	−5700***	−67.893***	−24.044***	−3200***	−92.882***	–	–	–
Pedroni	–	–	−81.654***	–	–	−82.877***	−67.304***	–
Westerlund (some)	–	–	–	–	–	–	–	−11.709***
Westerlund (all)	–	–	–	–	–	–	–	−10.689***

Note: ***, **, and * denotes rejection of the null hypothesis at the 1, 5, and 10% level of significance, respectively. The optimum lag order of the test is determined according to the Bayesian Information Criterion. The alternative hypothesis of the Kao and the Pedroni tests is that the variables are cointegrated in all panels. In the fisrt version of the Westerlund test, the alternative hypothesis is that variables are cointegrated in some of the panels. In the second version of the Westerlund test, the alternative hypothesis is that variables are cointegrated in all panels.

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TABLE 7 Long-run estimates on all counties.

Dependent variable: p	TWFE (1)	MG (2)	AMG (3)	CCEMG (4)	DCCEMG (5)	Sys_GMM (6)	SVR (7)	Panel SVR (8)
Mean-group estimates								
Real house price (p_{t-1})	0.993***	1.029***	0.928***	1.053***	1.117***	0.993***	0.998***	0.957***
	(0.001)	(0.002)	(0.002)	(0.006)	(0.001)	(0.000)	(0.001)	(0.024)
Real aggregate income (y_{t-1})	0.008***	0.062***	0.011***	0.020	0.012	0.017***	0.001	0.002**
	(0.001)	(0.005)	(0.003)	(0.014)	(0.021)	(0.007)	(0.012)	(0.000)
Housing units (u_{t-1})	0.002	−0.104***	−0.021*	0.031	0.025	−0.002	0.000	0.000
	(0.002)	(0.018)	(0.012)	(0.047)	(0.092)	(0.007)	(0.000)	(0.000)
Real yield spread (s_{t-1})	0.219***	−0.001**	0.002***	0.000	−1.658*	−0.000	0.001	0.001*
	(0.003)	(0.000)	(0.000)	(0.000)	(0.947)	(0.000)	(0.000)	(0.001)
Real oil prices (oil_{t-1})	−1.728***	−0.003***	0.000	0.000	0.036	−0.006***	−0.002**	−0.003**
	(0.023)	(0.000)	(0.000)	(0.000)	(0.359)	(0.000)	(0.000)	(0.002)
Real mortgage interest rate (r_{t-1})	−0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.003)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Real construction costs (c_{t-1})	−0.002	−0.006	0.016***	0.019	0.078***	−0.005*	−5.910	0.124*
	(0.003)	(0.004)	(0.003)	(0.017)	(0.025)	(0.003)	(6.327)	(0.068)
COVID-19 dummy	1.135***	0.003***	0.001*	0.008***	0.007***	0.003***	0.001**	0.000
	(0.015)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Common dynamic process	–	–	0.387***	–	–	–	–	–
			(0.012)					

(Continues)

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TABLE 7 (Continued)

Dependent variable: p	TWFE (1)	MG (2)	AMG (3)	CCEMG (4)	DCCEMG (5)	Sys_GMM (6)	SVR (7)	Panel SVR (8)
Constant	5.194***	−0.178	0.785***	2.720***	−0.419	−0.089***	0.002**	0.003**
	(0.072)	(0.202)	(0.143)	(0.468)	(0.862)	(0.033)	(0.001)	(0.001)
Residual diagnostics								
	IPS test	*CIPS test*	*CIPS test*	*CIPS test*	*CIPS test*	*IPS test*	*IPS test*	*CIPS test*
1 lag	−100***	−2.81***	−5.65***	−1.11	−2.30***	−150***	−11.35***	−10.27***
2 lags	−120***	−3.08***	−5.95***	−1.28*	−2.54***	−120***	−10.68***	−9.58***
3 lags	−53.41***	−1.91*	−4.68***	−1.24	−2.52***	−80***	−9.87***	−8.47***
4 lags	−54.45***	−1.91*	−4.78***	−1.22	−2.52***	−81***	−8.74***	−7.54***
Swamy test	3.56	66.15***	30.61***	26.73***	14.68***	28.46***	17.48***	15.87***
Aver. cross-correlation (CDS test)	−38.11	1711***	1424***	1832***	797***	25.86	51***	43***
Arellano–Bond test AR(1)	–	–	–	–		−6.03***	–	–
Arellano–Bond test AR(2)	–	–	–	–		−17.45***	–	–
Hansen test	–	–	–	–		43	–	–
Standard errors	Clustered	Jackknife	Huber weighting	Jackknife	Jackknife	Windmeijer estimator	Bootstrap	Bootstrap
Observations/cross-sections	115,939/431	115,939/431	115,939/431	115,939/431	115,939/431	115,939/431	115,939	115,939/431
Time period	January 2000–May 2022 (269 monthly observations)							

Note: *, **, and *** denote significance at the 10, 5, and 1% level, respectively. The Swamy test is a standardized version of Swamy's test for slope homogeneity proposed by Peasaran and Yamagata (2008) under the null hypothesis of weakly cross-sectional dependence of residuals. The CDS test stands for the cross-sectional dependence test of Pesaran (2015) under the null hypothesis of cross-section independence. With $n = 431$ and $T = 269$, the 1, 5, and 10% the critical values of the CIPS test with intercept are −2.15, −2.07, and −2.02, respectively (Pesaran, 2007). We use the typical Im–Pesaran–Shin test of homogenous coefficients for the TWFE, sys_GMM, and the SVR models. The selected lag length for cross-sectional means in the DCCEMG estimation is six according to $\sqrt[3]{269} \simeq 6$ (Chudik and Pesaran, 2015). Standard errors in parenthesis. We follow a two-stage GMM estimation process to account for possible endogeneity. The Arellano–Bond test for AR(2) in first differences rejects the null hypothesis for the existence of autocorrelation at the second lag of the residuals. The Hansen test of overidentifying restrictions fails to reject the null hypothesis that all instruments are valid.

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levels) for the lagged house prices, income, housing units, interest rate yield, oil prices, mortgage rate, construction costs, and a dummy variable that captures the effect of the COVID-19 pandemic period on housing market. The residual tests considered include the Im–Pesaran–Shin (IPS) and the cross-sectional IPS (Pesaran, 2007) tests of stationarity for the homogenous and the heterogenous panel estimators respectively, the Swamy test on slope homogeneity (Pesaran and Yamagata, 2008), and the cross-dependence sectional test (CDS) of Pesaran (2015). All coefficients are reported along with their robust standard errors. From the family of ML estimators, we report the coefficients for the typical SVR and the panel SVR models coupled with the linear kernel, where an affine structure is tractable. Standard errors for the SVR and the panel SVR models are derived with a bootstrapped repetition of 400 repetitions at the county level, based on the Politis and Romano (1994) stationary bootstrap procedure.

At the simplest TWFE model, the lagged house prices, income, interest rate yield and the COVID-19 dummy variable are statistically significant, with the largest effect reported on the oil price coefficient. The reported signs on all coefficients are as predicted by economic theory (Lai & Van Order, 2017; Oikarinen & Engblom, 2016; Oikarinen et al, 2018 for more detail on the signs). More specifically, an increase in income and in yield spread (sign of an expanding economy) are both expected to increase house prices. After all, the housing market represents 16.7% of total GDP,[11] which is a substantial part of the economy. Moreover, the lagged house prices coefficient is below one, suggesting that the model is stable.

The reported effect of oil is negative and of great magnitude as well as the COVID-19 dummy estimator; the model reports a 113.5% increase during the covid pandemic period on house prices that is not observed in our sample.[12] The aforementioned puzzles are solved, when we examine the residuals. Although the IPS test rejects the null hypothesis of nonstationarity, both the Swamy and the CDS test cannot reject the null hypothesis of remaining cross-sectional dependence in the residuals; the model does not entirely capture the cross-sectional dependence and thus all coefficients are biased.

Column (2) reports the MG estimator results considering cross-sectional dependency and spatial heterogeneity, introducing different coefficients per cross-section (county). The reported values refer to mean-group estimates. The real aggregate income is statistically significant and positive as expected and its effect has increased by almost eight times in comparison with the income coefficient of the TWFE model. The variable for housing units is now negative as expected and significant, while the effect of the term spread, oil prices and COVID-19 dummy are notably smaller, although not all of the correct signs (term spread has a negative coefficient). Again, the null hypothesis of nonstationarity in residuals is rejected; however, no remaining cross-sectional dependence is detected according to the Swamy and the CDS test. The AMG estimator reported in column (3) introduces a latent variable, grasping cross-dependence as a common dynamic process, but without examining lagged latent effects. The lagged house prices variable reported with the AMG estimator is below 1, the majority of coefficients have the correct sign and COVID-19 dummy exhibits no effect on house prices. The coefficients provided by the CCEMG estimator closely follow the AMG ones, but according to the CIPS test the estimator fails to provide the necessary stationary residuals.

In column (5), we report the coefficients for the DCCEMG; a dynamic version of the CCEMG model that uses lagged cross-sectional averages to account for cross-sectional dependence. The

[11] https://sgp.fas.org/crs/misc/IF11327.pdf

[12] The actual average increase reported in our sample during the COVID-19 period is 18.5% in comparison with the pre-COVID period.

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residuals are stationary, while all cross-sectional dependence is captured by the coefficients of the model. As we observe, the value of the lagged house prices is above one, suggesting an exploding behavior of the model and the yield spread coefficient has a large negative value that is not expected by theory. The aforementioned issues could be attributed to the addition of extra variables (that aim to remove cross-sectional dependence) with regard to the fundamental long-term relation (2) that affects coefficients' super-consistency. Hence, some of the attractive robustness features that usually come with super-consistent estimation features under cointegration are lost (Pedroni, 2007). Under this vein, the AMG estimator is considered superior to the DCEEMG. Controlling for the endogeneity of the lagged house prices, the system GMM model of column (6) exhibits interesting characteristics. The residuals are stationary, without remaining cross-sectional dependence, while all coefficients have the correct sign. More specifically, the most significant effect on house prices is observed in income and oil prices, followed by construction costs and the COVID-19 dummy variable, making the homogenous assumption rather restrictive.

Comparing the SVR and the SVR panel models, we observe that the first provides estimates that are close to the system GMM estimates. In contrast, the estimates of the panel SVR estimator are closer to the AMG, while the construction cost sign is now correctly identified. Interestingly, the coefficient of the COVID-19 dummy variable is not significant in the long-term, suggesting that the fluctuations during the COVID-19 pandemic has only short-run effects on house prices. Both models provide stationary residuals, without remaining cross-sectional dependence. Overall, from Table 7, we conclude that the AMG provides consistent long-run estimates superior to the other econometric estimators, although failing to provide the correct sign to all coefficients, while the panel SVR model provides estimates that follow the projections of economic theory.

After determining the long-run estimates, in Table 8, we proceed with the short-run estimates (in first-differences) of the error-correction model that measure the marginal effect of each variable on house prices. Results show that the nonstationarity hypothesis of residuals is rejected based on the (C)IPS tests for al estimators, while both the Swamy and the CDS test report no remaining cross-sectional dependence. The error term is statistically significant, has the expected negative sign and lies between −1 and zero in all, but the DCCEMG model. Moreover, the TWFE estimates are inconsistent as the reported lagged prices coefficient is above unity, suggesting an exploding autoregressive behavior.

For the rest but TWFE and DCCEMG models, the adjustment speed ranges from 2% for the SVR and the CCEMG models to 3% for the MG model, and reaches as high as 9.9% for the panel SVR and 30.8% for the system GMM model. The relevant literature reports an adjustment speed of around 5% (Holly et al., 2010; Oikarinen et al, 2018). It should be noted that higher adjustment rates toward mean values do not necessarily mean that one model is superior to the other, since mean reversion in error correction models is considered in infinite periods and it merely provides an estimate of the sensitivity of the model in exogenous transitory shocks. Finally, the momentum effect captured by the lagged prices coefficient is higher in the homogeneous (TWFE, sys_GMM, SVR) than the heterogenous (AMG, MG, CCEMG, panel SVR) models. Given that the relevant literature reports a momentum effect of around 25% (Oikarinen et al, 2018), the panel SVR corrects more efficiently for heterogeneity and cross-sectional dependence, reporting a momentum parameter of 30.7%.

On the examination of the expected signs according to theory, we expect a positive sign on income, yield spread, construction cost and the COVID-19 dummy and a negative sign on house units, oil prices, mortgage rate, and the error-correction term. Our empirical findings suggest that the model with correct marginal effects on house prices is the panel SVR model (column 8), exhibiting statistically significant coefficients for all but oil prices and the COVID-19 dummy,

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TABLE 8 Short-run estimates on all counties.

Dependent variable: Δp	TWFE (1)	MG (2)	AMG (3)	CCEMG (4)	DCCEMG (5)	Sys_GMM (6)	SVR (7)	Panel SVR (8)
Mean-group estimates	1.147***	0.759***	0.861***	0.722***	0.808***	0.938***	0.620***	0.307***
Real house price growth (Δp_{t-1})	(0.018)	(0.010)	(0.004)	(0.118)	(0.127)	(0.016)	(0.019)	(0.001)
Real aggregate income growth (Δy_{t-1})	0.034***	−0.212***	−0.036***	0.081**	0.116**	−0.226***	−0.171***	0.051**
	(0.008)	(0.007)	(0.003)	(0.034)	(0.055)	(0.009)	(0.058)	(0.024)
Housing units growth (Δu_{t-1})	0.015	0.152	−0.046**	0.323	0.575*	0.075***	−0.008***	−0.051***
	(0.013)	(0.118)	(0.023)	(0.201)	(0.303)	(0.017)	(0.002)	(0.047)
Real yield spread growth (Δs_{t-1})	0.062	0.001***	0.003***	0.000	0.000	0.001***	−0.002***	0.092***
	(0.254)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.034)
Real oil prices growth (Δoil_{t-1})	0.347	−0.015***	−0.006***	0.000	0.000	−0.018***	−0.057***	−0.061
	(0.247)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.165)
Real mortgage interest rate growth (Δr_{t-1})	−0.000	−0.000	0.000	−0.001	−0.001	−0.001***	−0.003***	−0.014***
	(0.001)	(0.011)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.006)
Real construction costs growth (Δc_{t-1})	0.012**	0.019	0.023***	−0.001	0.298***	0.084***	0.068***	0.114***
	(0.005)	(0.011)	(0.003)	(0.041)	(0.071)	(0.008)	(0.002)	(0.024)
Error-correction term $(p - p^*)_{t-1}$	−0.039***	−0.030***	−0.999***	−0.019***	0.000	−0.308***	−0.017***	−0.099***
	(0.027)	(0.007)	(0.002)	(0.004)	(0.000)	(0.014)	(0.001)	(0.002)
COVID-19 dummy	−0.019	0.000	0.003***	−0.000	0.001*	−0.001*	0.001*	0.001
	(0.247)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)
Common dynamic process	–	–	0.699***		–	–	–	–
			(0.023)					
Constant	0.012	0.000	−0.001***	−0.314	0.005***	0.000	0.001**	0.016
	(0.857*	(0.000)	(0.023)	(0.042)	(0.000)	(0.000)	(0.000)	(0.098)

(Continues)

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TABLE 8 (Continued)

Residual diagnostics								
	IPS test	*CIPS test*	*CIPS test*	*CIPS test*	*CIPS test*	*IPS test*	*IPS test*	*CIPS test*
1 lag	−200***	−6.09***	−6.18***	−5.39***	−5.405***	−230***	−154***	−97***
2 lags	−270***	−5.86***	−6.17***	−4.69***	−4.37***	−210***	−148***	−68***
3 lags	−170***	−5.12***	−5.99***	−3.39***	−3.27***	−140***	−124***	−48***
4 lags	−120***		−5.93***	−2.94***	−3.00***	−110***	−25***	−19***
Swamy test	69.86***	66.15***	8.19***	141.29***	159.26***	16.95***	17.85***	24.85***
Aver. cross-correlation (CDS test)	−6.12***	−251***	−1030***	2319***	2009***	−2843***	−248***	−235***
Arellano–Bond test AR(1)	–	–	–	–	–	−9.88***	–	–
Arellano–Bond test AR(2)	–	–	–	–	–	−5.14	–	–
Hansen test	–	–	–	–	–	−0.07	–	–
Standard errors	Clustered	Jackknife	Huber weighting	Jackknife	Jackknife	Windmeijer estimator	Bootstrap	Bootstrap
Observations /cross-sections	115,007/431	115,007/431	115,007/431	115,007/431	115,007/431	115,007/431	115,007/431	115,007/431
Time period	February 2000–May 2022 (268 monthly observations)							

Note: *, **, and *** denote significance at the 10, 5, and 1% level, respectively. The Swamy test is a standardized version of Swamy's test for slope homogeneity proposed by Peasaran and Yamagata (2008) under the null hypothesis of weakly cross-sectional dependence of residuals. The CDS test stands for the cross-sectional dependence test of Pesaran (2015) under the null hypothesis of cross-section independence. With $n = 431$ and $T = 269$, the 1, 5, and 10% the critical values of the CIPS test with intercept are −2.15, −2.07, and −2.02, respectively (Pesaran, 2007). We use the typical Im–Pesaran–Shin test of homogenous coefficients for the TWFE, sys_GMM, and the SVR models. The selected lag length for cross-sectional means in the DCCEMG estimation is six according to $\sqrt[3]{268} \simeq 6$ (Chudik and Pesaran, 2015). Standard errors in parenthesis. We follow a two-stage GMM estimation process to account for possible endogeneity. The Arellano–Bond test for AR(2) in first differences rejects the null hypothesis for the existence of autocorrelation at the second lag of the residuals. The Hansen test of overidentifying restrictions fails to reject the null hypothesis that all instruments are valid.

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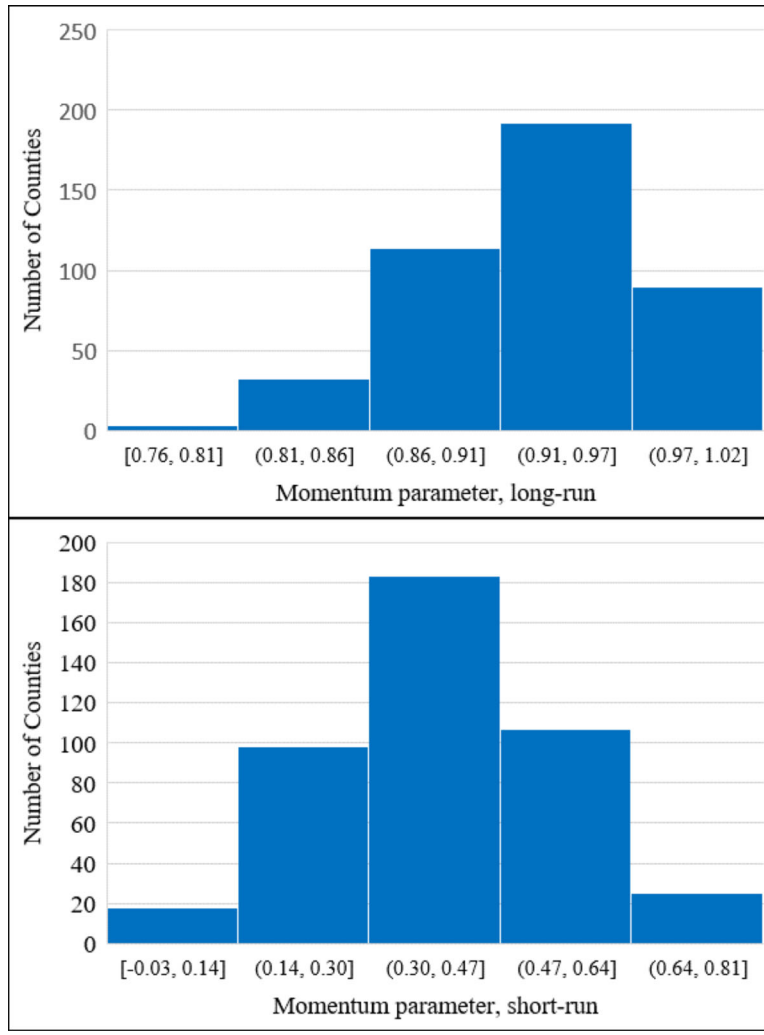

FIGURE 2 Heterogenous short- and long-term momentum parameter frequencies per number of counties. [Color figure can be viewed at wileyonlinelibrary.com]

while it attributes the correct sign to all coefficients. According to the panel SVR model, the highest marginal effect originates from construction costs with an average of 11.4% across sections, followed by the yield spread at 9.2%, income at 5.1%, housing units' growth with −5.1%, and mortgage rates with −1.4%. Overall, in terms of sign and magnitude the coefficients of the panel SVR model adhere closer to literature than other estimates. To depict spatial heterogeneity of the momentum effect on house prices, we report in Figure 2 the distribution of the respective individual coefficients per county.

The first subplot of Figure 2 shows that most counties exhibit a momentum parameter value in the range of 0.91–0.97, followed by 0.86–0.91 and 0.97–1.02. The picture is different in the second subplot that depicts the short-run momentum parameters, with the majority of coefficients in the range of 0.30–0.47 and 0.47–0.64. This difference suggests that long-run house prices tend to be more "sticky"; prices do not trend away from long-run equilibrium levels, while the marginal effect of exogenous short-term fluctuations is small. In Figure 3, we extend the heterogeneity of

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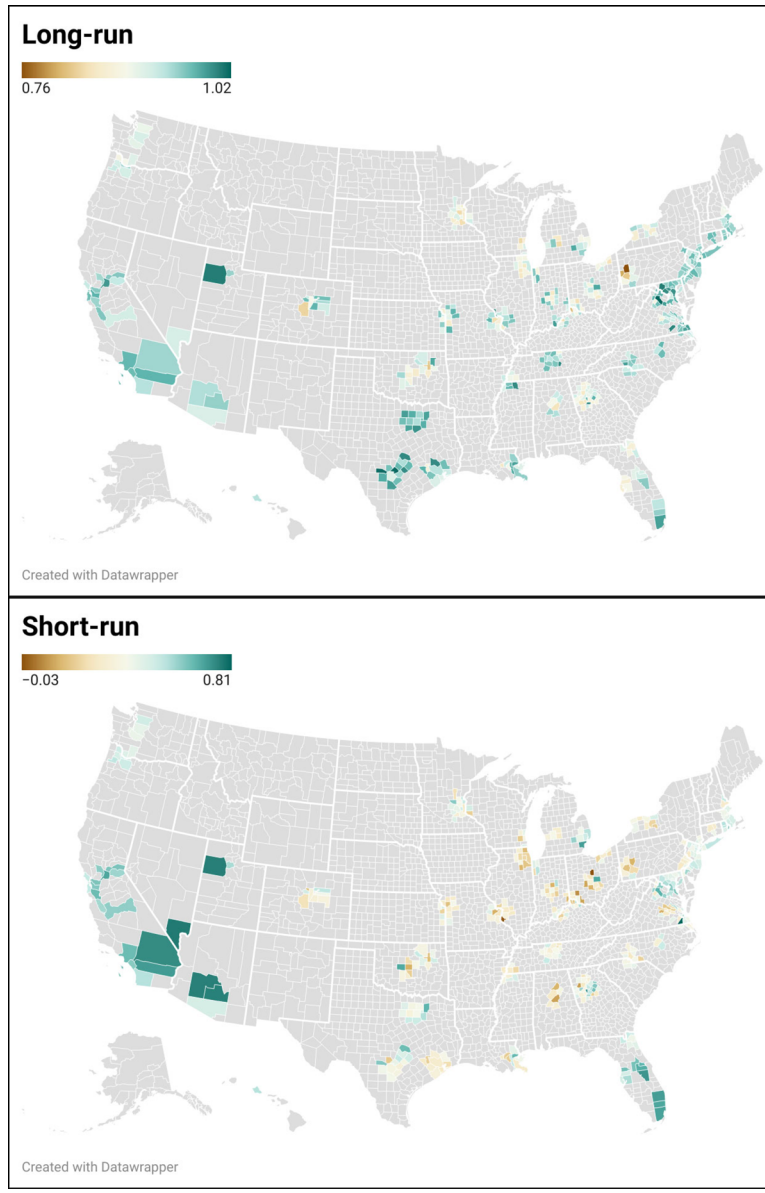

FIGURE 3 Geographic distribution of short- and long-term momentum parameters. [Color figure can be viewed at wileyonlinelibrary.com]

coefficients to their spatial distribution, matching parameters to counties.

The spatial distribution of momentum parameters also reveals the significant heterogeneity between counties. Starting from the first subplot of long-run coefficients, we observe that the highest values are observed in the west, south-west, and east counties as opposed to mid-west counties. This discrepancy is more intense in the short-run coefficients plot (second subplot), where west, south-west, and east regions have a much larger momentum parameter in comparison with inland counties. The geographic distribution of momentum elasticities generally conforms with expectations, with higher elasticities found mostly in coastal locations and following the pattern of income

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distribution and housing supply.[13] Higher income and limited housing supply provide a tendency for increased housing demand that keeps the housing market stable and with lower fluctuations.

Our analysis insofar suggests a significant momentum coefficient in the long-run, but a smaller one in the short-run equation. The panel SVR model correctly detects the signs of all coefficients as expected by economic theory and provides coefficient values that are closer to empirical practice. The elements of spatial heterogeneity and cross-sectional dependence are intense in our housing prices dataset, calling for a treatment in the forecasting exercise to accurately grasp the data generating process in local housing markets.

3.2 | Forecasting house prices

In the previous section, we examine the ability of the panel SVR method to describe house price dynamics at the county level, in tandem with popular panel econometric estimators that account for heterogeneity. Nonetheless, we do not reach to analytical expressions for the other ML methods, while we do not fully exploit the merits of a panel SVR; a ML model oriented in forecasting (Mullainathan & Spiess, 2017). To this end, we proceed with forecasting house prices at the county level, based on a rolling window approach of 60 monthly observations, with a sliding window of 1 month, measuring the RMSE at in-sample and out-of-sample forecasting. In order to get robust inferences of the forecasting ability of the estimator, we evaluate the forecasting performance of all methods in different samples; either as a single panel of 431 counties or independently at the MSA level, training on 50 individual panel datasets (six MSAs have only one county).

All models are built according to the error-correction model structure and we consider alternative forecasting horizons of 1, 3, 6, 12, and 24 months ahead. Naturally, residential construction comes with a significant lag as real estate investments are typically long-term investments. Thus, most business-oriented applications adhere to forecasts of annual or longer horizons. However, house prices are also used in higher frequencies up to the monthly one as predictors of business cycles (Arias et al., 2016), determinants of real economic activity (Cesa-Bianchi et al., 2015) and instruments in hedging inflation (Christou et al., 2018), suggesting that short-term forecasts of house prices are of interest according to the relevant. To account for either end of the problem, we consider both short and long forecasting horizons, to access the forecasting performance in situations where the data generating process varies as the time horizon increases.

In Panel A of Table 9, we report in-sample RMSE per forecasting horizon for the entre panel dataset. As a benchmark we report the forecasting performance of a RW model. The RW model assumes market efficiency; the best forecast is the last observed value. Outperforming the RW model suggests that the market is not efficient and house prices can be consistently predicted. Results in Panel A show that all but the TWFE model (in some horizons) outperform the benchmark RW in in-sample forecasting. The most accurate model is the XGBoost that achieves an average RMSE of 0.056 in all horizons, followed by the RF with an average forecasting error of 0.177. The most accurate econometric model is the AMG that achieves the third lowest average error of 0.416, presumably due to the ability to model heterogeneity with a lower number of coefficients in comparison with CCEMG and DCCEMG. The panel SVR model improves its forecasting performance at larger forecasting horizons, but achieves the 4th lowest average forecasting error with a RMSE of 0.579.

[13] https://www.census.gov/library/visualizations/interactive/2019-median-household-income.html and https://www.census.gov/library/visualizations/interactive/bps-new-privately-owned-housing-unit-authorizations.html

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TABLE 9 Forecasting accuracy at the county level in an entire panel.

Horizon (months)	1 (1)		3 (2)		6 (3)		12 (4)		24 (5)		Average (6)	
Panel A: In-sample RMSE												
RW	1.087	(13)	0.902	(12)	0.910	(12)	0.998	(12)	1.086	(13)	0.974	(12)
TWFE	0.734	(9)	2.441	(13)	1.412	(13)	1.206	(13)	0.735	(9)	1.306	(13)
MG	0.641	(6)	0.671	(6)	0.653	(6)	0.651	(6)	0.641	(5)	0.651	(6)
AMG	0.425	(3)	0.412	(3)	0.409	(3)	0.409	(3)	0.425	(3)	0.416	(3)
CCEMG	0.722	(8)	0.752	(8)	0.749	(8)	0.729	(8)	0.722	(8)	0.735	(8)
DCCEMG	0.716	(7)	0.709	(7)	0.740	(7)	0.727	(7)	0.716	(7)	0.722	(7)
Sys_GMM	0.921	(12)	0.876	(11)	0.804	(10)	0.938	(11)	0.922	(12)	0.892	(11)
LASSO	0.756	(10)	0.758	(9)	0.762	(9)	0.770	(9)	0.786	(10)	0.766	(9)
ANN	0.824	(1)	0.836	(10)	0.825	(11)	0.844	(10)	0.855	(11)	0.837	(10)
RF	0.135	(2)	0.181	(2)	0.182	(2)	0.188	(2)	0.197	(2)	0.177	(2)
SVR	0.585	(4)	0.630	(5)	0.645	(5)	0.643	(5)	0.664	(6)	0.633	(5)
XGBoost	0.056	(1)	0.056	(1)	0.056	(1)	0.056	(1)	0.057	(1)	0.056	(1)
SVR panel	0.618	(5)	0.606	(4)	0.579	(4)	0.541	(4)	0.553	(4)	0.579	(4)
Panel B: Out-of-sample RMSE												
RW	1.022	(12)	0.879	(10)	0.884	(11)	0.963	(11)	1.022	(11)	0.954	(11)
TWFE	2.951	(13)	4.739	(13)	3.040	(13)	2.943	(13)	2.951	(13)	3.325	(13)
MG	0.789	(7)	0.795	(5)	0.790	(4)	0.781	(5)	0.789	(5)	0.789	(5)
AMG	0.928	(11)	0.914	(12)	0.911	(12)	1.112	(12)	1.228	(12)	1.019	(12)
CCEMG	0.789	(8)	0.795	(6)	0.790	(5)	0.781	(4)	0.789	(4)	0.789	(4)
DCCEMG	0.685	(4)	0.701	(2)	0.677	(2)	0.639	(2)	0.687	(2)	0.678	(2)
Sys_GMM	0.929	(10)	0.886	(11)	0.804	(6)	0.931	(10)	0.930	(9)	0.896	(10)
LASSO	0.773	(6)	0.770	(3)	0.767	(3)	0.752	(3)	0.755	(3)	0.763	(3)
ANN	0.836	(9)	0.853	(8)	0.831	(8)	0.827	(6)	0.821	(6)	0.834	(8)
RF	0.595	(2)	0.790	(4)	0.815	(7)	0.862	(7)	0.889	(8)	0.790	(6)
SVR	0.713	(5)	0.872	(9)	0.871	(10)	0.926	(9)	0.944	(10)	0.865	(9)
XGBoost	0.617	(3)	0.832	(7)	0.830	(9)	0.867	(8)	0.864	(7)	0.802	(7)
SVR panel	0.296	(1)	0.267	(1)	0.241	(1)	0.218	(1)	0.240	(1)	0.252	(1)

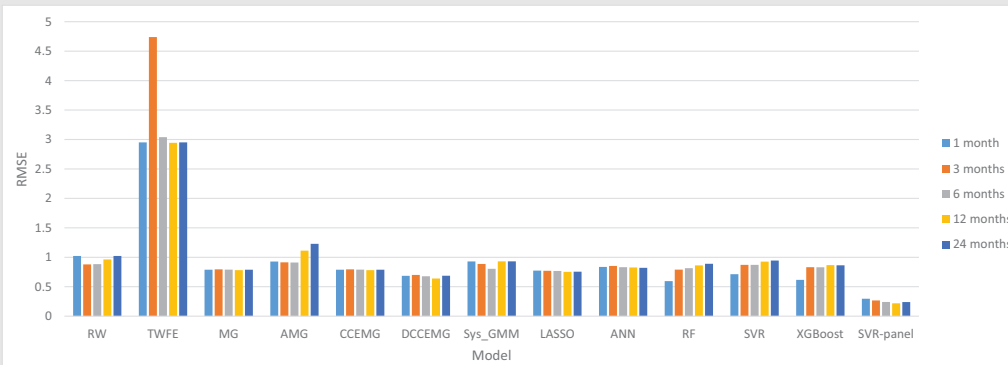

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TABLE 9 (Continued)

The table reports RMSE for the forecasting exercise where we treat the entire set of counties as one panel-dataset of 431 cross-sections. All RMSE are multiplied with 100 to increase readability. In parentheses the order of each model per horizon according to the forecasting error. The Diebold and Mariano (1995) test suggests that the panel SVR model outperforms all models in out-of-sample forecasting, at the 1% level of significance. All statistics are available at the Supplementary File. The bar chart that accompanies the table refers to out-of-sample RMSE values.

Nevertheless, the true forecasting performance of any method can only be measured by forecasting unknown data during training. In Panel B, we report the out-of-sample RMSEs, where the panel SVR model is the most accurate, achieving half or less of the forecasting error of the second-best RF for the monthly horizon (0.296 vs. 0.595, respectively) and DCEEMG for the rest of horizons (panel SVR exhibits an average error of 0.252 while DCEEMG an average error of 0.678). The TWFE model and for some horizons the AMG perform worse than the RW model, while the XGBoost exhibits significantly higher forecast errors (mean value 0.802) in comparison with its in-sample performance (0.056), suggesting overfitting the data during training. This finding contrasts our previous section in describing house prices dynamics, where we highlight the ability of AMG to adhere to the true data generating process, corroborating to the findings of Oikarinen et al (2018) who also support AMG in describing local housing dynamics. Presumably, AMG has the ability to grasp heterogeneity but fails to generalize adequately.[14] The empirical findings suggest that the panel SVR model outperforms all models in out-of-sample forecasting at the 1% level of significance. Moreover, in comparison with the typical SVR model, the inclusion of spatial dependence in the kernel function greatly improves the forecasting performance, due to the construction of an alternate dimensional space in comparison with the one created with the typical SVR model.

Overall, the panel SVR estimator outperforms all competing techniques in out-of-sample forecasting, while we do not find evidence that any other method consistently outperforms it in all horizons, regardless of using of either homogenous or heterogenous coefficients, econometric or ML estimators, and controlling or not for cross-sectional dependence. Thus, a simple addition of heterogenous coefficients and cross-sectional correlations to the popular SVR model greatly improves its forecasting performance, including all merits of the econometric literature on a well-performing ML estimator.

For robustness test, we evaluate the accuracy of all methods in forecasting average county house prices in 50 smaller panels, trained individually at the MSA level, since six MSAs have only one county and are not panel datasets. In Table 10, we report the average in-sample and out-of-sample RMSEs over all MSAs, while the detailed results for each MSA dataset are available in the Supplementary File.

As expected, the decrease in the number of observations with the individual MSA panel datasets increases the forecasting error, especially for the TWFE and the system GMM methods. The empirical findings do not deviate from those of Table 9. The most accurate model in in-sample forecasting is again the XGBoost (Panel A) with an average error across horizons of 0.310. The RF and the panel SVR models are in the second and third place in terms of accuracy, with 0.885 and 1.082 mean RMSEs across horizons, respectively. The most profound difference is the significant decline in the accuracy of the TWFE and the system GMM models, who consistently underperform the RW model and in a magnitude up to hundreds. We reach similar conclusions for the

[14] In the Supplementary File, we test the null hypothesis of equal forecasting error of the panel SVR model over all other methods with the Diebold and Mariano (1995) test.

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TABLE 10 Forecasting accuracy averaged per individual MSA.

Horizon (months)	1		3		6		12		24		Average	
	(1)		(2)		(3)		(4)		(5)		(6)	
Panel A: In-sample RMSE												
RW	2.987	(6)	4.489	(11)	4.596	(10)	5.148	(10)	5.740	(11)	4.592	(11)
TWFE	47.306	(12)	35.261	(12)	24.555	(12)	34.497	(12)	46.328	(12)	37.589	(12)
MG	3.649	(8)	3.407	(6)	3.312	(6)	3.311	(6)	3.260	(5)	3.388	(5)
AMG	1.234	(4)	1.549	(4)	1.531	(4)	1.517	(4)	1.576	(4)	1.481	(4)
CCEMG	3.979	(10)	3.865	(9)	4.858	(11)	5.443	(11)	3.600	(7)	4.349	(10)
DCCEMG	4.114	(11)	4.029	(10)	4.137	(9)	4.049	(9)	3.821	(8)	4.030	(9)
Sys_GMM	156.341	(13)	165.742	(13)	290.143	(13)	213.195	(13)	241.558	(13)	213.396	(13)
LASSO	3.722	(9)	3.738	(7)	3.765	(7)	3.820	(7)	3.960	(9)	3.801	(7)
ANN	3.458	(7)	3.765	(8)	3.822	(8)	3.946	(8)	4.138	(10)	3.826	(8)
RF	0.775	(3)	0.894	(2)	0.895	(2)	0.915	(2)	0.946	(2)	0.885	(2)
SVR	2.950	(5)	3.146	(5)	3.171	(5)	3.222	(5)	3.291	(6)	3.156	(6)
XGBoost	0.307	(1)	0.308	(1)	0.311	(1)	0.309	(1)	0.314	(1)	0.310	(1)
SVR panel	0.749	(2)	1.142	(3)	1.146	(3)	1.185	(3)	1.187	(3)	1.082	(3)
Panel B: Out-of-sample RMSE												
RW	2.594	(2)	4.048	(10)	4.144	(10)	4.567	(10)	4.929	(11)	4.056	(10)
TWFE	68.925	(12)	57.680	(12)	43.808	(12)	53.212	(12)	63.942	(12)	57.513	(12)
MG	3.787	(9)	3.789	(7)	3.777	(8)	3.732	(8)	3.757	(7)	3.768	(7)
AMG	3.241	(5)	3.557	(4)	3.541	(4)	3.527	(3)	3.586	(4)	3.490	(5)
CCEMG	3.787	(8)	3.789	(8)	3.777	(6)	3.732	(7)	3.757	(6)	3.768	(8)
DCCEMG	3.828	(10)	3.766	(5)	3.692	(5)	3.558	(4)	3.533	(3)	3.675	(4)
Sys_GMM	156.414	(13)	166.417	(13)	290.570	(13)	212.649	(13)	240.727	(13)	213.355	(13)
LASSO	3.769	(7)	3.770	(6)	3.777	(7)	3.724	(6)	3.761	(8)	3.760	(6)
ANN	4.364	(11)	4.492	(11)	4.575	(11)	4.637	(11)	4.657	(10)	4.545	(11)
RF	2.628	(3)	3.257	(2)	3.320	(2)	3.500	(2)	3.411	(2)	3.223	(2)
SVR	3.417	(6)	3.794	(9)	3.881	(9)	3.871	(9)	3.958	(9)	3.784	(9)
XGBoost	2.812	(4)	3.442	(3)	3.495	(3)	3.650	(5)	3.615	(5)	3.403	(3)
SVR panel	0.706	(1)	1.103	(1)	1.122	(1)	1.136	(1)	1.139	(1)	1.041	(1)

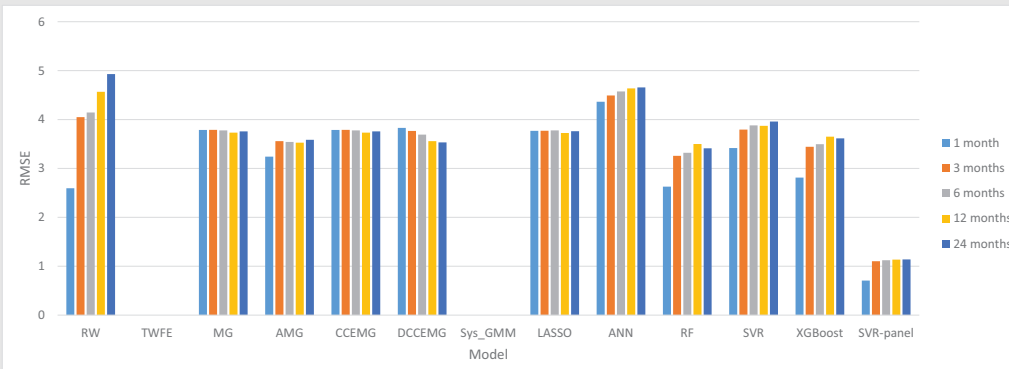

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TABLE 10 (Continued)

The table reports average RMSE over MSAs, for the exercise that treats each MSA as a separate dataset and MSA's respective counties are the cross-sections of each panel. All RMSEs are multiplied with 100 to increase readability. In parentheses the order of each model per horizon according to the forecasting error. The Diebold and Mariano (1995) test suggests that the panel SVR model outperforms all models in out-of-sample forecasting, at the 1% level of significance. All statistics are available at the supplementary file. The bar chart that accompanies the table refers to out-of-sample RMSE values. Values for TWFE and sys-GMM are suppressed to increase readability.

out-of-sample forecasts in Panel B. The panel SVR estimator has the highest generalization ability with an average error of 1.041 across horizons, followed again by the RW with three times higher average RMSE of 3.223 across horizons and the XGBoost with an error of 3.403. The TWFE and the system GMM models exhibit low generalization ability, suggesting that in the econometric context the heterogenous treatment of coefficients has an obvious advantage in forecasting, while again no specific pattern emerges in a comparison between econometric and ML estimators. In the Supplementary File, we report Diebold and Mariano (1995) statistics on the superiority of the panel SVR over all other estimators in out-of-sample forecasting at the 1% level of significance, at the MSA level. In the same Supplementary File, we evaluate the performance of all estimators in forecasting house prices at the ZIP level for the 56 MSAs of our sample, compiling a panel dataset of 6988 cross-sections (ZIP codes) over 268 monthly observations (growth rates). Unfortunately, the specific exercise is an autoregressive one, due to data availability restrictions on all other control variables at such a detailed level. The empirical findings are quantitative similar, with the panel SVR estimator exhibiting the lowest out-of-sample forecasting error, outperforming all econometric and ML models.

Overall, our forecasting exercises suggest that the panel SVR method exhibits the highest generalization ability with statistical significance, whereas the XGBoost method tends to overfit the data. Given that all forecasts refer to rolling window averages, the results are robust to the existence of structural breaks and sample specific characteristics. The RF estimator exhibits high forecasting ability, in terms of both in-sample and out-of-sample errors. In the Supplementary File, we report an evaluation of variable importance in RF forecasting, suggesting that lagged house prices are more important at shorter horizons, while the role of oil prices and interest rate spread becomes more significant at longer horizons. Reported results corroborate to the selection of the independent variables of our error-correction models proposed by theory and empirical practice. We do not detect an overall superiority of ML estimators over econometric ones and the performance of any method increases with the number of cross-sections.

3.3 | House prices bubble identification

While forecasting house prices is of keen interest to market participants in shaping regional trends of the housing market, the formation of house price bubbles is closely monitored by all monetary authorities. Toward this end, we evaluate the ability of the panel SVR estimator to foresee bubble formations in house prices. As Stiglitz (1990) argues, a bubble refers to house price levels that depart substantially from their "fundamental" value. In a similar content, Lind (2009) postulates that house price bubbles are dramatic increases in house prices that are succeeded by rapid falls, but the optimum growth speed and scale of the rise and drop that defines a bubble is hard to be identified. Although there is broad accordance among researchers that MSAs differ in their likelihood to experience house price bubbles (see among others Shiller, 2014; Glaeser & Nathanson,

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2017), most studies do not include in the estimation process heterogenous region responses during the formation phase.

We proceed with the "stiglitzian" approach and define a bubble as a deviation of house prices from their fundamental value, relative to ex-post expectations of regression models (see among others Case & Schiller, 2003; Wheaton & Nechayev, 2008 and Bourassa et al., 2019). A crucial first step is the ex-ante determination of what defines the benchmark (ground truth) for the existence of a bubble. For that, we follow the price–rent ratio approach of Case and Schiller (2003), where deviations of the price–rent ratio from its long-term average imply the existence of a bubble. The (logarithmic) price–rent ratio is an easy applicable index that adheres closely to the price dynamics of the real estate market and often exhibits higher accuracy than the price-income ratio, the growth rate of house prices or univariate regression models (Bourassa et al., 2019). For robustness check, in the Supplementary File, we perform a comparison of periods identified as bubbles with the price–rent approach to periods identified with the Phillips and Shi (2019) method. The latter consists of augmented Dickey–Fuller regressions on rolling windows to identify nonstationary behavior of timeseries, using bootstraps to estimate the empirical distribution of the tests. Using different similarity tests, we conclude that both methods identify almost the same bubble periods.

As postulated by the real estate economic theory, we consider the real aggregate income, the real mortgage interest rate and the real term spread as explanatory variables from the demand side and housing units, construction costs, oil prices, mortgage rates, and term spread from the supply side. We also consider the lagged value of the dependent variable on the right-hand side of the equation to cater for potential feedback effects in housing demand. In Figure 4, we depict the (logarithmic) standardized price–rent ratios for the 6 most populated and the six less populated MSAs. The standardization lies in dividing each ratio by its long-term average of the respective MSA over the entire period of the sample. In this way, given that both the rent and the house prices are expressed in logarithmic form, we get a percentage deviation of the ratio from its long-run (fundamental) value. The selection of the MSA level in Figure 4 is motivated by the need to increase the readability of the plots as our sample covers 431 counties, since we keep the county level to the bubble identification exercise. In the vein of Case and Shiller (2003),[15] we select a 20% deviation threshold for identifying a bubble, depicted with the horizontal straight line.

By visually inspecting Figure 4, we infer that price bubbles exhibit variations both in terms of time of appearance as well as magnitude. For instance, Chicago and Washington D.C. exhibit higher deviations from their long-run ratios in comparison with Dallas and Houston. All bubbles deflate around 2009, with Dallas and Houston prolonging the deflation period up to 2011. A similar picture is depicted in the smaller six MSAs (bottom subplot), where Honolulu barely surpasses the 20% threshold for a period of only 4 years, Fresno has a small 6-year bubble period and Rochester and Tulsa exhibit a quasi-linear trend with negative slope. In the Supplementary File, we depict the respective plots for the rest MSAs, that are not included in Figure 4. Overall, we observe significant heterogeneity in bubble formation between MSAs, suggesting that the heterogenous estimators could provide more accurate forecasts than the homogenous econometric and ML.

We exploit house prices forecasts from the previous forecasting exercise, where we forecasted house prices 1, 3, 6, 12, and 24 months ahead, based on a rolling window scheme. The forecasted values are standardized (divided) with their respective average long-term values. A deviation higher than the 20% threshold[16] is considered an indication for the formation of a bubble. Thus,

[15] In the Supplementary File, we examine alternative thresholds of 0, 10, and 30%.

[16] Alternative thresholds are reported in the Supplementary File with quantitative similar results.

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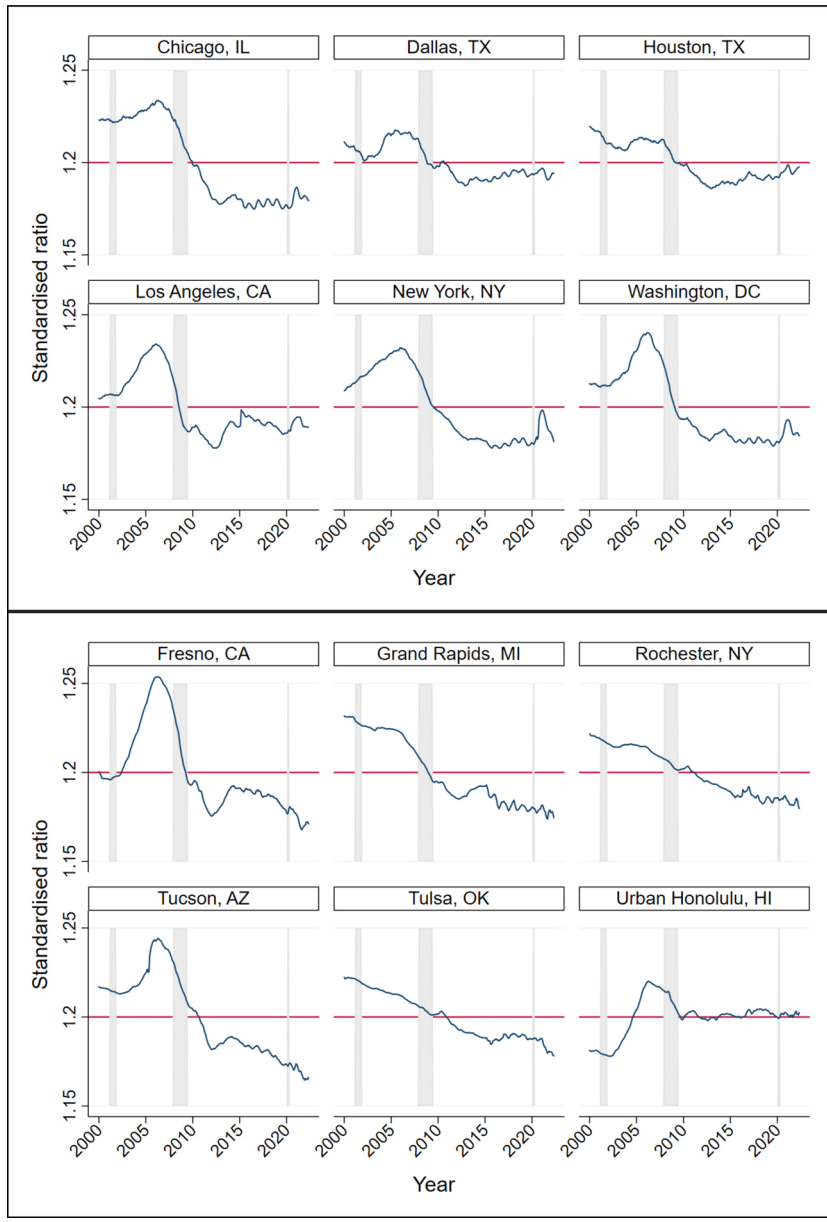

FIGURE 4 Standardized price–rent ratios for the larger six (top) and smallest six (bottom) MSAs in our sample by population. The horizontal line expresses deviations from the 20% threshold that defines a bubble. Grey areas are U.S. recessions. [Color figure can be viewed at wileyonlinelibrary.com]

we evaluate the forecasting accuracy of the forecast models by comparing the identified to the benchmark; the bubble periods identified by the price–rent ratio. In Table 11, we report the balanced (weighted) accuracy of all models using the ratio of bubble against nonbubble periods as weight on the correctly identified periods, given that our dataset is unbalanced toward nonbubbly periods. Moreover, since our interest lies with the identification of bubbles and not tranquil

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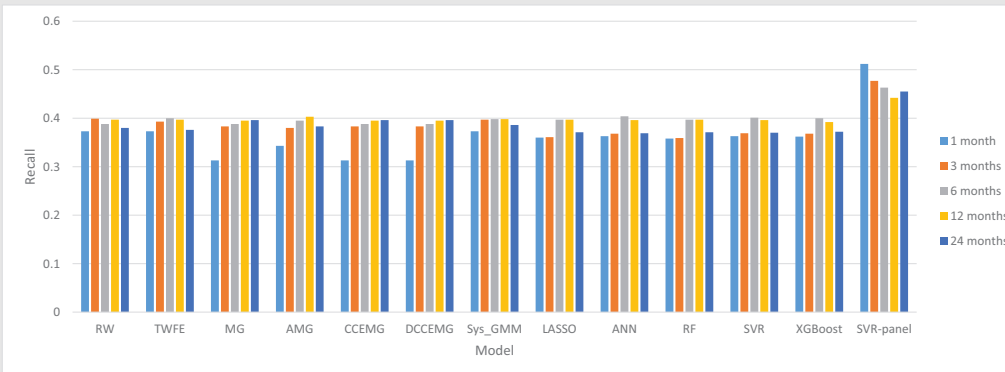
TABLE 11 Bubble Identification (threshold 0.20).

Horizon (months)	1		3		6		12		24		Average	
	(1)		(2)		(3)		(4)		(5)		(6)	
Panel A: Balanced accuracy												
RW	0.443	(9)	0.447	(8)	0.437	(9)	0.441	(9)	0.424	(9)	0.438	(9)
TWFE	0.475	(4)	0.432	(9)	0.459	(8)	0.464	(8)	0.433	(8)	0.453	(8)
MG	0.282	(11)	0.302	(11)	0.303	(11)	0.306	(11)	0.310	(11)	0.301	(11)
AMG	0.307	(10)	0.318	(10)	0.326	(10)	0.325	(10)	0.319	(10)	0.319	(10)
CCEMG	0.282	(12)	0.302	(12)	0.303	(12)	0.306	(12)	0.310	(12)	0.301	(12)
DCCEMG	0.282	(13)	0.302	(13)	0.303	(13)	0.306	(13)	0.310	(13)	0.301	(13)
Sys_GMM	0.470	(5)	0.477	(4)	0.464	(7)	0.465	(7)	0.480	(6)	0.471	(6)
LASSO	0.481	(2)	0.469	(5)	0.551	(2)	0.544	(2)	0.525	(2)	0.514	(2)
ANN	0.453	(7)	0.455	(7)	0.471	(6)	0.491	(6)	0.468	(7)	0.468	(7)
RF	0.449	(8)	0.460	(6)	0.517	(4)	0.521	(3)	0.506	(4)	0.491	(5)
XGBoost	0.477	(3)	0.492	(3)	0.500	(5)	0.510	(5)	0.493	(5)	0.494	(4)
SVR	0.462	(6)	0.499	(2)	0.520	(3)	0.520	(4)	0.517	(3)	0.504	(3)
SVR panel	0.556	(1)	0.544	(1)	0.576	(1)	0.588	(1)	0.595	(1)	0.572	(1)
Panel B: Recall												
RW	0.373	(2)	0.399	(2)	0.388	(10)	0.397	(4)	0.380	(7)	0.387	(4)
TWFE	0.373	(3)	0.393	(4)	0.400	(4)	0.397	(5)	0.376	(8)	0.388	(3)
MG	0.313	(11)	0.383	(5)	0.388	(11)	0.395	(11)	0.396	(2)	0.375	(11)
AMG	0.343	(10)	0.380	(8)	0.395	(9)	0.403	(2)	0.383	(6)	0.381	(5)
CCEMG	0.313	(12)	0.383	(6)	0.388	(12)	0.395	(12)	0.396	(3)	0.375	(12)
DCCEMG	0.313	(13)	0.383	(7)	0.388	(13)	0.395	(13)	0.396	(4)	0.375	(13)
Sys_GMM	0.373	(4)	0.397	(3)	0.398	(6)	0.398	(3)	0.386	(5)	0.390	(2)
LASSO	0.360	(8)	0.361	(12)	0.397	(8)	0.397	(6)	0.371	(10)	0.377	(9)
ANN	0.363	(5)	0.368	(10)	0.404	(2)	0.396	(8)	0.369	(13)	0.380	(6)
RF	0.358	(9)	0.359	(13)	0.397	(7)	0.397	(7)	0.371	(11)	0.376	(10)
XGBoost	0.363	(6)	0.369	(9)	0.401	(3)	0.396	(8)	0.370	(12)	0.380	(7)
SVR	0.362	(7)	0.368	(11)	0.400	(5)	0.392	(9)	0.372	(9)	0.379	(8)
SVR panel	0.512	(1)	0.477	(1)	0.463	(1)	0.442	(1)	0.455	(1)	0.470	(1)



The table reports recall and balanced accuracy for identifying bubbles over all MSAs, for the exercise that treats all data as one panel with the counties being the cross-sections. In parentheses the rank of each model per horizon according to its accuracy. The bar chart that accompanies the table refers to recall values.

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periods, we also report recall; the actual percentage of bubble periods identified on the entire sample.[17]

We find that the model with the highest average forecasting accuracy over all periods (Panel A, column 6) is the panel SVR, achieving an accuracy of 0.572, followed by the LASSO model with 0.514, SVR with 0.504 and XGBoost with 0.494. Nevertheless, when individual horizons are taken into account, apart from the panel SVR model, no other exhibits the same forecasting performance in all horizons. For instance, the LASSO estimator that is the second most accurate on the monthly horizon, goes to fifth place at the quarterly and ranks second at the semi-annual, annual and bi-annual horizons, respectively. The RF that exhibits high accuracy in house pricing forecasting, ranks eighth at the monthly horizon, sixth at the quarterly, fourth at the semi-annual, third at the annual, and fourth at the bi-annual, respectively. An interesting finding is that all four heterogenous econometric estimators (MG, AMG, CCEMG, and DCCEMG) are ranked last in terms of accuracy and below the RW model. In fact, MG, CCEMG, and DCCEMG exhibit the same forecasting performance, while the TWFE and the system GMM consistently perform better. Presumably, the addition of the mean cross-sections to cater for spatial heterogeneity increases the volatility of forecasts (Pesaran, 2007).

On bubble forecasting (excluding nonbubble periods) using recall (Panel B), we reach to different conclusions. Starting our analysis from the individual horizons, apart from the panel SVR, all other models' forecasts are irrelevant at the monthly and quarterly horizons, since they underperform the RW. In the semi-annual horizon, the ANN and the XGBoost exhibit the second and the third rank with 0.404 and 0.401, respectively, while the panel SVR achieves 0.463. In the annual horizon, the panel SVR has a recall of 0.442, the AMG a recall of 0.403 and the system GMM 0.398, while in the bi-annual horizon the three heterogenous econometric estimators achieve a recall of 0.396, just behind the panel SVR. Overall, the average recall over all horizons reported in column (6) suggests that the most accurate model is the panel SVR with a recall of 0.470, followed by the system GMM with a recall of 0.390, the TWFE model with 0.388 and the RW with 0.387, that renters all other models of no practical use. In the Supplementary File, we report the forecasting accuracy of all models on alternative thresholds with quantitative similar results, as well as the performance of the panel SVR model at the MSA level, considering each MSA as a distinct panel dataset.

Our empirical findings corroborate those of Plakandaras et al. (2015), since we are able to forecast successfully house price bubbles almost 2 years before they form, although we use a more granular dataset (county level) and of higher frequency that their annual dataset measured at the national level. Moreover, our study controls for heterogeneity and cross-sectional dependence, an element largely missed by all previous studies. In the Supplementary File, we repeat the forecasting exercise at the lowest available ZIP level with quantitative similar results, although we base our analysis on autoregressive models. Bourassa et al. (2019) also examine the ability of various methods to identify house price bubbles for Chicago, Miami and San Francisco and three European MSAs, concluding that regression methods have poor ability in detecting bubbles as they form. Nevertheless, their sample is limited, highly aggregated, and potentially misses cross-sectional dependence as bubbles are highly contagious between local house markets (see Figure 3 for an exposition of spatial clustering in momentum effects). Fabozzi and Xiao (2019) also detect bubble formation in eight U.S. cities improving on the augmented Dickey–Fuller regressions of Phillips and Shi (2019), missing the cross-sectional dependence characteristic as they examine each city

[17] Recall is denoted as $\frac{\text{True Positives}}{\text{True Positives+False Negatives}}$.

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independently. Overall, our study is the first to study bubble detection under spatial heterogeneity and cross-sectional dependence, exploiting the advantages of ML in forecasting at the detailed county level.

3.4 | Economic evaluation

While forecasting superiority in terms of statistical loss functions is important, the true interest of any market participant lies in the ability to use forecasts in shaping profitable trading strategies. To provide economic evaluation of the proposed forecasting framework, we compile a monthly dataset of stock returns from residential REITs covering the period January 2000 to May 2022. The economic evaluation follows each model's forecast on future REIT stock returns, which are formalized to a decision to invest in REIT stocks or buy 3-month Treasury bills, considered as the risk-free alternative. The selection of residential REIT stocks stems from the fact that the determining factors of a successful investment in residential buildings follow the same principles of supply and demand of the residential housing market examined in the previous sections, and are heavily dependent on house prices that is the focus of our study. An alternative path would be to measure the economic significance of the proposed forecasting methodology by evaluating individual house re-sales and their impact on the housing market, but we are bounded by data availability. We leave this strand of research for the future.

Examining our sample, we find that many REITs invest in more than one state. Thus, we estimate the return from an investment in REIT stocks per state based on an equally weighted portfolio that includes all REITs that are active in the specific state for the 40 states reported in Table 2. The choice for equally weighted portfolios follows Jacobs et al. (2014) who show that equally weighted portfolios tend to outperform value-weighted ones in out-of-sample forecasting exercises.

We train forecasting models based on our 60-month rolling window training scheme examined earlier to keep everything comparable and forecast 1, 3, 6, 12, and 24 months ahead. Following Wang et al. (2020), a mean-variance investor allocates her wealth every month at each state between the REIT stock portfolio and the risk-free asset based on the weight $\hat{w}_{t+1}$ given by:

$$\hat{w}_{t+1} = \frac{1}{g}\frac{\hat{r}_{t+1}}{\hat{\sigma}^2_{t+1}}, \tag{23}$$

that leads to returns in period $t+1$

$$R_{t+1} = \hat{w}_{t+1} * r_{t+1} + (1 - \hat{w}_{t+1}) * rf_{t+1}, \tag{24}$$

where R_{t+1} are the actual returns, r_{t+1} the actual returns from the REIT portfolio, $\hat{r}_{t+1}$ the forecasted returns from the REIT portfolio by the evaluated model, rf_{t+1} the actual risk-free returns, and $\hat{\sigma}^2_{t+1}$ the t+1 variance of the REIT portfolio defined by a 5-year rolling window (see Rapach et al., 2016). The risk-averse parameter g is set to 3, as in Rapach et al. (2016) and Campbell and Thompson (2008).

To create a measure of economic significance that is independent of the forecasted period, we also measure the certainty equivalent return (CER), which is the risk-free rate that the investor is willing to accept instead of differentiating between REIT stocks and the risk-free asset in (24),

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TABLE 12 Economic evaluation.

Horizon (months)	1	3	6	12	24	Average	
	(1)	(2)	(3)	(4)	(5)	(6)	
Panel A: Annualized returns (annualized Sharpe ratios in parenthesis)							
RW	0.026*	0.033	0.000	−0.004	0.015	0.014	(11)
	(0.063)	(0.574)	(−0.675)	(−0.929)	(−0.094)		
TWFE	0.026	0.013	0.033	0.006	0.028	0.021	(8)
	(−0.079)	(−0.239)	(0.628)	(−0.419)	(0.339)		
MG	0.012	0.007	0.008	0.008	0.007	0.008	(13)
	(0.367)	(0.042)	(0.145)	(0.216)	(0.047)		
AMG	0.014	0.014	0.014	0.014	0.015	0.014	(10)
	(0.710)	(0.710)	(0.707)	(0.710)	(0.699)		
CCEMG	0.034**	0.037**	0.033**	0.034**	0.032**	0.034	(6)
	(0.724)	(0.988)	(0.693)	(0.956)	(0.726)		
DCCEMG	0.013	0.011	0.012	0.009	0.008	0.011	(12)
	(0.343)	(0.173)	(0.260)	(0.074)	(0.048)		
System GMM	0.017	0.018	0.019	0.025	0.016	0.019	(9)
	(−0.229)	(−0.227)	(−0.275)	(−0.196)	(−0.017)		
LASSO	0.044	0.044	0.043	0.044	0.057	0.046	(4)
	(−0.612)	(−0.574)	(−0.639)	(−0.565)	(−0.124)		
ANN	0.077**	0.005	0.014	0.004	0.011	0.022	(7)
	(0.611)	(−0.182)	(0.125)	(−0.265)	(0.009)		
RF	0.057	0.054	0.019	0.048*	0.050	0.046	(3)
	(−0.432)	(−0.307)	(−1.289)	(−0.051)	(−0.664)		
XGBoost	0.054	0.094	0.016	0.033	0.030	0.045	(5)
	(−0.516)	(0.899)	(−1.260)	(−0.564)	(−0.192)		
SVR	0.067**	0.044	0.071**	0.075**	0.071**	0.066	(2)
	(0.442)	(−0.212)	(0.622)	(0.682)	(0.676)		
Panel SVR	0.137***	0.128***	0.129***	0.131***	0.131***	0.131	(1)
	(0.885)	(1.301)	(0.919)	(0.863)	(0.876)		
Panel B: Certainty equivalent returns							
RW	0.060	0.075	0.000	0.000	0.011	0.022	(12)
TWFE	0.062	0.006	0.076	0.000	0.063	0.041	(7)
MG	0.027	0.017	0.018	0.019	0.015	0.019	(13)
AMG	0.031	0.031	0.031	0.031	0.031	0.031	(8)
CCEMG	0.078	0.086	0.077	0.079	0.072	0.078	(4)
DCCEMG	0.029	0.024	0.027	0.020	0.015	0.023	(10)
System GMM	0.017	0.018	0.034	0.062	0.008	0.028	(9)
LASSO	0.060	0.056	0.061	0.062	0.117	0.071	(6)
ANN	0.050	0.000	0.032	0.003	0.023	0.022	(11)

(Continues)

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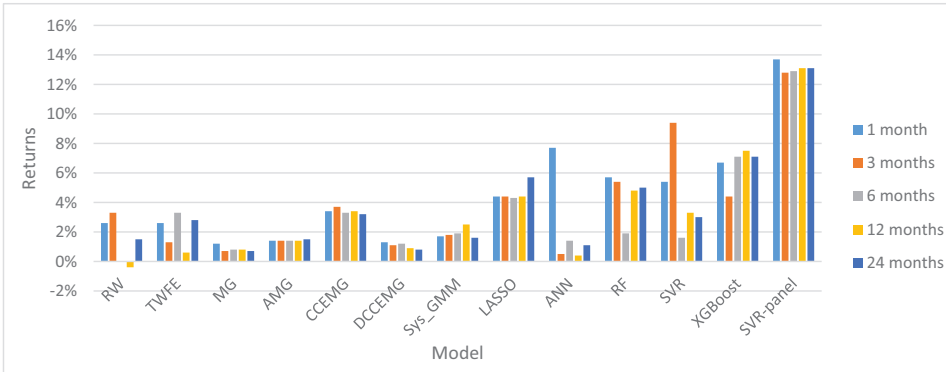
TABLE 12 (Continued)

Panel B: Certainty equivalent returns							
RF	0.122	0.106	0.000	0.050	0.104	0.076	(5)
XGBoost	0.113	0.200	0.000	0.023	0.062	0.080	(3)
SVR	0.135	0.057	0.160	0.167	0.139	0.132	(2)
Panel SVR	0.339	0.316	0.317	0.318	0.308	0.320	(1)



The table reports the statistical and economic forecasting performance of all models over the panel dataset where the states are cross-sections. *, **, and *** indicate significance at the 10, 5, and 1% levels, respectively. Asterisks are presented for the Sharpe ratio (GRS test). In parentheses of the last column at the right of the table is the rank of each model per horizon according to its performance. The bar chart that accompanies the table refers to annualized returns.

depending on the mean return μ_R and the variance σ_R^2 of the actual returns.

$$\text{CER} = \mu_R - \frac{1}{2} * g * \sigma_R^2 \tag{25}$$

All Sharpe ratio (SR) values are based on annualized returns. The significance of the economic performance is determined with the Gibbons et al. (1989) test, which evaluates the null hypothesis of zero return based on the F-distribution of the SR as follows:

$$\frac{T\,(T-N-1)}{N\,(T-2)} Q \sim F_{N,T-N-1,a}, \tag{26}$$

where $Q = \left(\frac{\sqrt{1+\widehat{\text{SR}}^2}}{\sqrt{1+\overline{\text{SR}}^2}}\right)^2 - 1$

with $\widehat{\text{SR}}$ denoting the SR of the forecasted returns, $\overline{\text{SR}}$ is the SR of the historical average REIT returns per state, N denotes the number of assets, and a the level of statistical significance. A rejection of the null hypothesis implies higher financial performance of a diversified portfolio in comparison with investing in the REIT stocks portfolio only. In Panel A of Table 12, we report the annualized returns and the SRs, while CER are reported in Panel B.

As we observe in Panel A, most SRs are not statistically significant. Only the CCEMG, the SVR, and the panel SVR achieve statistically significant SRs. The panel SVR model achieves the highest annualized returns across horizons, with an average of 13.1%. The second most profitable strategy is the one based on the forecasts of the SVR model with an average of 6.6%, followed by the LASSO and the RF models. The most profitable strategy based on an econometric model is the CCEMG,

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with an average of 3.4%. Apparently, the panel SVR model shapes a strategy with twice the return in comparison with other methods and significantly higher SRs. For the CER-based evaluation (Panel B), the interest rate reported using the panel SVR model is 32%, significantly higher than the 13.2% of the SVR, the 8% of the XGBoost, and the 7.8% of the CCEMG. For both panels, the trading strategies based on the ML methods are significantly more profitable than their econometric counterparts. In the Supplementary File, we report annualized returns SRs and CERs at the MSA level for the panel SVR model.

Our empirical findings compare with Bianchi and Guidolin (2014) who forecast all REITs returns traded on the U.S. stock market based on Markov switching vector autoregression models for the period January 1972 to December 2012. The authors report annualized returns of 12.6% and CER of 5.7% that are well below our estimated returns and CER, presumably due to the limiting flexibility of the linear form of the models considered. Unfortunately, all other empirical studies in REIT returns use a composite index of REIT stocks and do not account for the entirety of the REIT market under spatial heterogeneity.

4 | CONCLUSION

The economic theory and the voluminous empirical practice have highlighted the importance of forecasting house prices at the local level, given that house markets have strong regional characteristics. Nevertheless, spatial heterogeneity and cross-sectional dependence are rarely considered in the relevant literature and most studies on house price dynamics are based solely on homogenous (across sections) panel estimators.

In this article, we argue that local characteristics of the housing market are of key importance to describing house price dynamics, increasing the adherence to the local data generating mechanism as well as to the forecasting efficiency. In doing so, we follow the recent upsurge in the use of ML methods in economic forecasting and introduce heterogeneity and cross-correlations to the popular SVR methodology. The empirical findings suggest that the panel SVR models adhere closely to the most efficient econometric panel estimators in the field and exhibit exceptional forecasting performance. In order to evaluate the potential of the proposed method, we compile different panel datasets for the 56 most populated U.S. MSAs at the county level and evaluate the ability of the method to forecast house prices at the local level, identify bubbles in the housing market as they form and measure the economic significance of forecasts for market participants.

Based on an error-correction formulation, the panel SVR model correctly identifies the expected effect on a pool of independent variables that express the demand and supply forces of the housing market at the local level, both in the short- and the long-run equation, corroborating to the empirical literature and the assumptions of urban economic theory. The respective econometric estimators are in most cases biased, either due to overlooking heterogeneity, cross-dependence, or the introduction of many additional parameters to control for the aforementioned effects that create multicollinearity issues.

Furthermore, we evaluate the forecasting performance of a plethora of novel econometric and popular ML estimators in house prices forecasting at the county level. We follow two different approaches exploiting the entire panel dataset of 431 counties over the period February 2000 to May 2002 and alternatively on multiple panel estimation at the MSA level. The empirical findings of either approach suggest that the panel SVR model outperforms all other estimators in terms of out-of-sample RMSE across various forecasting horizons, while the second most accurate method is the econometric DCCEMG, followed by the LASSO estimator. At the individual MSA level, the

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panel SVR retains its superiority with the second accurate model being a RF model. Moreover, we do not detect an overall superiority of the other ML methods over econometric ones, while the performance of the proposed method increases with the number of cross-sections.

While house prices are of keen interest to market participants, the recent experience from the great recession of 2008 suggests that bubbles in the housing market can result in severe economic ramifications. Extending the forecasting exercise to bubble identification, we argue that the proposed methodology can identify bubbles as they form both timely and with relevant success. Focusing solely on the percentage of actual bubble periods correctly identified, our empirical findings suggest that the panel SVR model achieves a recall of 0.470 over the 0.390 of the system GMM, 0.388 of the TWFE and 0.388 of the RW model. From a portfolio formation perspective, the SVR model can achieve a mean return of 13.1% and a CER of 32% that is twice as high than any other model considered in this article and higher from the relevant values reported in the literature.

Our work has multiple policy implications spanning from the market participant, who seeks to grasp local trends and reach to positive investment returns, to the monetary policy official that is interested in bubble formations at the local housing market, to avoid contagion effects to the national real estate market. Future research paths could extend the forecasting exercise to the individual house sale level and potentially other economic phenomena that also exhibit high spatial heterogeneity.

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